

The Benefits and Costs of a Food Waste Tax and its Implications for Climate Change Mitigation

Seunghoon Lee*

2024-12-02

Abstract

Lifecycle greenhouse gas (GHG) emissions from wasted food rival those from road transport, and waste management is one of the largest budget items for municipalities worldwide. While levying a corrective tax on food waste is a textbook approach to address these externalities, evidence of its benefits and costs remains scarce. Exploiting plausibly exogenous expansions of a unit-based food waste tax in South Korea, I report three findings. First, the tax substantially reduces food waste by discouraging excessive grocery purchases. Second, households sustain food intake despite buying less, but at the cost of additional time spent on meal production. Third, the tax's benefits far outweigh its costs because the tax promotes more efficient food use, which not only reduces externalities but also offsets households' additional time costs.

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*Department of Economics, University of Missouri (seunghoon.lee@missouri.edu). I am deeply grateful to Ryan Kellogg, Koichiro Ito, and Dan Black for their guidance and mentorship. I also thank Kathy Baylis, Nathan Chan, Tatyana Deryugina, Eyal Frank, Don Fullerton, Matthew Gibson, Edward Jaenicke, Amir Jina, Namrata Kala, Taeho Kim, Christopher Knittel, Shanjun Li, Joshua Linn, Jacquelyn Pless, Michael Price, Devesh Raval, Brian Roe, Nicholas Ryan, Frank Schilbach, Hee Kwon Seo, Jeffrey Shrader, Richard Sweeney, Rebecca Taylor, Derek Wu, Eric Zou, and seminar participants at the AAEA National Conference, AERE summer conference, ASSA Annual Meeting, APPAM fall conference, DEEP workshop, IPWSD workshop, MIT, Monash University, National University of Singapore, OWSEET workshop, Penn State University, Seoul National University, Sogang University, the University of Chicago, University of Connecticut, University of Illinois Urbana Champaign, University of Mannheim, University of Missouri, for their helpful comments. Sangjin Kim at the Ministry of Food, Agriculture, Forestry and Fisheries in South Korea graciously offered to help with data acquisition. All errors are mine.

1 Introduction

Globally, one-third of all food is discarded, generating lifecycle greenhouse gases (GHGs) comparable to the emissions from the entire road transport sector (Gustavsson et al. 2011, IPCC 2014, FAO 2015). Further, waste management is often the largest municipal budget item, diverting resources from more productive investments (Kaza et al. 2018).¹ Consumer-end food waste is particularly problematic due to its high volume—responsible for up to 60% of the total food waste in developed countries—and cumulative environmental impact along the supply chain (Gustavsson et al. 2011, EPA 2021). As such, reducing food waste at home has become a critical priority for policymakers (United Nations 2015, IPCC 2018, EPA 2021, OECD 2021).

Theoretically, a Pigouvian tax on food waste can make households internalize externalities and generate the socially optimal level of food waste (Pigou 1920).² Whether implementing such a tax enhances social welfare in practice, however, remains an empirical question, as it is unclear a priori which abatement strategies households will adopt, and these strategies have varying implications for welfare. For instance, while both spending time improving food storage practices and engaging in illegal dumping can reduce the tax burden, their impacts on environmental and fiscal externalities (i.e., benefits) and abatement efforts (i.e., costs) can be starkly different.

This paper provides the first empirical evidence on the welfare effect of a unit-based food waste tax (hereafter “unit tax”), which charges households based on the quantity of food waste they generate. To assess the benefits, I study the tax’s impact on households’ food purchasing, consumption, and waste, as well as the resulting changes in environmental and fiscal externalities. To evaluate the costs, I build on the insights from the household production model (Becker 1965), and investigate how the tax impacts households’ meal production time. Importantly, by investigating the tax’s impact across different stages of food use and time allocation, I can assess its overall welfare effects while explicitly accounting for households’ abatement strategies.

The empirical application exploits two waves of plausibly exogenous unit tax expansions in South Korea. Since 2005, households were required to separate food waste from landfill waste and pay

¹Solid waste management accounts for 4%, 11%, and 19% of total municipality budget in high-, middle-, and low-income countries, respectively (Kaza et al. 2018). For instance, NYC spent \$1.5 billion in 2021 alone for providing waste related services (NYC Council Finance Division 2024).

²An alternative design is to tax a market good associated with emissions (e.g., food) rather than directly taxing the emissions (e.g., food waste). However, this policy faces challenges in terms of technical feasibility, political acceptability, and efficacy, as discussed in Section 6.2.

a food waste tax. During this initial phase, most households were charged a flat tax, while some—depending on their region and housing type—were subject to the unit tax (Wave 1 expansion). Later, with mounting concerns over food waste driven GHG emissions and public finance burden, the central government mandated local governments to expand the unit tax to all households by 2013 (the Wave 2 expansion). However, as the physical infrastructure for unit tax collection can be costly, the unit tax rolled out over time, and 30% of households were still paying flat tax in 2017. The average tax rate after the Wave 2 expansion was 6 cents per kilogram of food waste, or \$1.3 per month for a household with average waste levels (\$1 = KRW 1,100 throughout the paper).

To leverage these variations, I collect four different datasets on purchased, consumed, and wasted food, as well as household time usage, and match them to the unit tax status based on time, region, and housing type. For identification, I compare treated (i.e., levied the unit tax) households to not-yet and never treated households using the two-way fixed effect approach. When aggregated data is used, I compare geographic units with varying levels of treatment intensity. I also show that the results are robust to alternative methods proposed in the recent difference-in-differences literature (Cengiz et al. 2019, Callaway and Sant’Anna 2021, Sun and Abraham 2021, Borusyak et al. 2022).

The empirical exercises produce three key results. First, I find that the policy is highly effective: for an average household, the tax reduces annual food waste and grocery purchases by 19% (46kg) and 5.5% (41kg), respectively. Comparing these effect in levels (41kg vs. 46kg) suggests that purchasing less food in the first place is the primary abatement strategy, while other measures, such as illegal dumping, account for only 10% of the observed reduction in food waste. Further, the policy effect is three times larger in magnitude for perishable items (fresh vegetables and fruits) than storable items, which is plausible given that perishable items are easier to become waste when not consumed in time. Importantly, these effects do not come at the cost of households’ nutritional needs: the tax does not reduce food and nutrient intake at home, suggesting that the reduction in grocery purchases comes from the previously wasted—rather than consumed—part of the food basket.

The estimated changes in food usage imply that the tax reduces lifecycle GHG emissions from wasted food by 3.1 million ton of CO₂eq annually at the national level—equivalent to the emissions from 750 thousand passenger vehicles. Additionally, reduced food waste lowers government spending on waste management, resulting in net savings of \$43 million per year from the unit tax, even after accounting for implementation costs. Beyond reducing these externalities, the tax generates a private

benefit as well by encouraging households to save \$169 annually on grocery expenditures.

Next, I estimate the abatement cost of reducing food waste (i.e., purchasing fewer groceries). Given that food intake does not decline after the tax, I empirically examine whether the tax leads to an increase in (1) time spent on meal production or (2) meal production productivity, building on the household production model where households combine time and groceries to produce meals (Becker 1965). I find that households spend 7.6% more time (55 additional hours per year) on meal production, incurring an annual time cost of \$175.³ Web search data further indicate that households use this extra time to organize and store food more efficiently. However, I find no evidence that the tax increases productivity.

Using these estimates, the final part of the paper performs a simple welfare analysis. I estimate that the unit tax reduces one ton of CO_2eq at a cost of \$23 and generates over \$500 million in net benefits annually. These effects are driven by the tax promoting more efficient food use, which not only mitigates environmental and fiscal externalities but also offsets households' additional time costs through savings on grocery expenditures.

Related literature. This paper contributes to three different bodies of literature. First, it provides the first empirical evidence on the welfare effects of a corrective tax on food waste. Despite the theoretical promise of such taxes, much of the existing food waste literature focuses on evaluating non-financial policies, such as nudges or informational campaigns (see reviews by Reynolds et al. (2019); National Academies of Sciences, Engineering, and Medicine (2020); and Zhang et al. (2023)). The limited research on food waste taxes remains predominantly theoretical (Hojgard et al. 2013, Katare et al. 2017, Hamilton and Richards 2019). By leveraging a unique nationwide policy experiment in South Korea and multiple datasets on food and time usage, this paper provides credible evidence that a food waste tax can significantly reduce environmental and fiscal externalities at moderate social costs. These findings offer a strong support for implementing market-based food waste policies.

Second, and more broadly, this paper departs from earlier studies on municipal solid waste (MSW) taxing (i.e., pricing) policies in two important ways.⁴ This is the first paper to explicitly estimate how upstream consumption changes in response to a downstream tax, contrasting with existing re-

³To express the time cost in monetary terms, I use the returns to shopping—money saved for additional time spent on shopping—as a proxy for the opportunity cost of time (Aguilar and Hurst 2007a).

⁴MSW is a broader category of waste than food waste, including not only food scraps but also packaging, clothing, bottles, newspapers, appliances, and other everyday materials.

search that primarily examines changes at the waste disposal stage (e.g., increased recycling) (Fullerton and Kinnaman 1996, Allers and Hoebein 2010, Carattini et al. 2018, Bueno and Valente 2019, Valente 2021). Investigating the upstream effect is important because better disposal can reduce only a small fraction of lifecycle environmental externalities from waste.⁵ Second, earlier studies underestimate the social cost of waste by an order of magnitude—even without considering the upstream effects—by abstracting away from the GHG emissions from waste.⁶ These two differences could explain why earlier studies in general conclude waste pricing policies are likely welfare-harming.

Lastly, this paper complements voluminous literature on the economics of climate change mitigation, which has predominantly studied conventional carbon-intensive sectors such as power, manufacturing, heating, or transportation (Fowlie et al. 2016, Andersson 2019, Gerarden et al. 2020, Ito and Zhang 2020, Reynaert 2021). In contrast, this paper focuses on food, a sector often understudied despite growing recognition of its critical role in achieving climate goals.⁷

The rest of the paper proceeds as follows. Section 2 provides background on the GHG emissions from food waste and waste policy changes in South Korea. Section 3 details the data sources and provides summary statistics. Section 4 and Section 5 estimate the benefits and costs of the tax, while Section 6 discusses welfare effects and additional policy implications. Section 7 concludes.

2 Background

2.1 Negative Externalities from Food Waste

Lifecycle greenhouse gas (GHG) emissions from wasted food are estimated at 4.4 GtCO₂eq per year, constituting 8% of total anthropogenic GHG emissions (FAO 2015). This is comparable to the annual emissions from the entire road transport at 5.1 GtCO₂eq (IPCC 2014). These substantial emissions can be attributed to two key factors. First, food production is highly carbon-intensive due to, for instance, land clearing (deforestation) and enteric fermentation in livestock, and nitrogen fertilizers (Springmann et al. 2018, IPCC 2019). Second, food waste generates large amounts of methane

⁵For food waste, which is the largest (44%) waste category in MSW (Kaza et al. 2018), the farm-to-kitchen stage are responsible for 90% of lifecycle GHG emissions (Crippa et al. 2021).

⁶For instance, Repetto et al. (1992) estimates that the social cost of waste is \$5/ton due to “air and water pollution, noise, and other disamenities”. However, when GHG emissions from food degradation in landfills are considered, the social cost of waste is at least \$207/ton at \$190/ton of the social cost of carbon (EPA 2016, 2023).

⁷Clark et al. (2020) finds that an immediate end of fossil fuel consumption is insufficient to meet the Paris Agreement’s 1.5°C climate goal without changing the world’s food system.

when it decays in landfill.⁸ Notably, the relative contribution of these two sources differs by an order of magnitude: Crippa et al. (2021) shows that 90% of the lifecycle GHGs from food waste are attributable to the farm-to-kitchen stage.

In addition to greenhouse gas (GHG) emissions, food waste imposes a significant fiscal externality. MSW management is often the largest budget item for many municipalities, accounting for 4%, 11%, and 19% of total municipality budget in high-, middle-, and low-income countries (Kaza et al. 2018). For example, New York City spent \$1.5 billion in 2021 alone on waste-related services (NYC Council Finance Division 2024). Similarly, South Korea allocated \$1 and \$0.5 billion to landfill and food waste management in 2015 (Ministry of Environment 2015a). Given that food waste constitutes the largest portion of global MSW at 44% (Kaza et al. 2018), reducing food waste could alleviate fiscal pressures and redirect public funds toward more productive areas worldwide.

In addition to the GHG emissions and government spending, food waste causes additional externalities such as biodiversity losses, soil degradation, water depletion, bad odor, urban space constraints throughout its lifecycle (EPA 2021). Hence, the GHG and fiscal cost represents a conservative estimate of the total social cost of wasted food.

Given these large externalities, reducing food waste has gained significant policy attention in recent years.⁹ While food waste prevention (the so-called “source reduction”) is deemed most desirable (as illustrated by the EPA’s food recovery hierarchy in Appendix Figure D.1), prevailing policies such as encouraging donation, composting, or energy recovery focus on recycling already generated wasted food (National Academies of Sciences, Engineering, and Medicine 2020). A corrective tax on food waste can be a powerful alternative to those existing policies, as it has the potential to prevent excessive food purchase in the first place.

2.2 Waste Policies in South Korea

Wave 0 (1995-2004): unit-based tax on landfill waste. Between 1970–90, waste quantity in South Korea increased by sevenfold. To address resulting landfill capacity problem, a national landfill tax was implemented in 1995. The policy had two key features. First, source-separated recycle items

⁸Indeed, landfill is the third largest methane source in the US despite widely adopted methane-to-energy facilities (EPA 2016, Kaza et al. 2018).

⁹For instance, United Nations Sustainable Development Goal 12.3 calls for halving per household food waste at the retail and consumer levels by 2030 (United Nations 2015). In the US, EPA and USDA adopted the target of cutting food waste at the retail and consumer level by 50% by 2030 (USDA and EPA 2021).



Figure 2.1: Unit Based Food Waste Tax Collection Methods

such as glass bottles were picked up free of charge. Second, for other MSW (including food waste), households were required to purchase and use an official garbage bag. Prior studies found that this policy reduced the amount of waste in landfill (Hong 1999).

Wave 1 (2005-2012): Food waste segregation and partial implementation of unit-based food waste tax. In response to increasing environmental concerns over food waste in landfill, the Ministry of Environment mandated that municipalities collect food waste separately and recycle it into compost or animal feed. Also, to offset the cost of food waste management, municipalities were permitted to impose a food waste tax on households. Since the Wave 1 policy focused on *segregation* rather than *reduction*, the collection system was strategically designed to minimize operational costs. Consequently, food waste from apartment complexes were typically collected without measurement and residents were charged a flat tax (\$1-2 per month).¹⁰ The collection methods for non-condo residents were more varied: about half of municipalities charged flat fees while the rest required households to purchase official food waste bags (Figure 2.1 (a)), which typically cost 1 cents/kg during this period. As detailed Section 5.2, the Wave 1 expansion is utilized to identify the time costs of the unit tax.

Wave 2 (2013-): Nation-wide expansion of the unit based food waste tax. With mounting concerns over GHG emissions from food waste, the central government mandated that municipalities expand the unit tax by 2013. As the Appendix Figure D.2 (a) shows, the overall fraction of households un-

¹⁰In this paper, “apartments” and “condominiums” are used interchangeably. Unlike in the U.S., where “apartment” typically refers to rental units, in South Korea, “apartment” refers to both rented and owner-occupied units.

der the unit tax was around 30% between 2009 and 2012 (during the Wave 1 period) but sharply increased to over 70% in 2017, with the most dramatic change occurring in 2013.¹¹ Further, the map of the Wave 2 expansion in Appendix Figure D.2 (b) reveals substantial variation in the expansion across different municipalities.

The expansion happened under two different collection regimes: typically, official trash bags for non-condo residents and smart card systems for condominium residents (Figure 2.1 (b)).¹² Importantly, because the smart card system is costly (approximately \$1,500 per kiosk), not all the households received the unit tax treatment at the same time. As such, about 30% of households did not get the treatment even in 2017, and these never-treated households form an important part of the control group in subsequent empirical exercises.

The average tax rate after the Wave 2 expansion was 6 cents per kilogram of food waste, or \$1.3 per month for a household with average waste levels. As the tax revenue covers only 30% of the food waste management costs, local governments end up spending substantial amount of public funds to provide these services—for instance, the City of Seoul spent \$0.15 billion in 2015 alone.

3 Data

Food waste quantity and tax policy. The Unit-Based Waste Policy Yearbook from the Ministry of Environment provides annual municipality-level data on waste quantities, tax rates, and tax regimes (flat or unit tax). As the food waste tax is applied to non-bulk generators, which includes both households and small restaurants, the data reflects waste quantities from both sources. To ensure comparability with other food usage datasets, I use a conversion ratio based on statistics from the City of Seoul, which indicate that the food waste quantity generated by one restaurant is, on average, equivalent to that of seven households. Based on this conversion, I calculate the per-household food waste by dividing the total food waste quantity by the combined number of residential households and restaurant-equivalent households.¹³ I restrict my analysis to the 63 municipalities in metropoli-

¹¹The figure is based on data from 63 municipalities in the metropolitan Seoul area, aligning with the geographic scope of the empirical analysis on food usage.

¹²To make two units (i.e., volume for bags and weight for smart card systems) comparable, I convert volume to weight using an official conversion ratio of 0.75kg/liter from the Ministry of Environment (Executive Order “2015-164”).

¹³The City of Seoul records food waste quantity separately for residential households and small restaurants since 2014. I use 2014 and 2015 data to calculate the conversion ratio. The data is available at https://stat.eseoul.go.kr/statHtml/statHtml.do?orgId=201&tblId=DT_201004_J070007&conn_path=I3.

tan Seoul area, which coincides with the geographic coverage of the grocery purchase data, from 2009 to 2015.¹⁴ I cross validate the tax status from the Yearbook using municipality ordinances.¹⁵ For the Wave 1 policy changes, which predates the Yearbook data, I leverage a commissioned study from the Ministry of Environment, which documents the tax status as of 2009 (Kim et al. 2010).

Grocery purchase. Grocery purchase information comes from consumer grocery panel data from the Rural Development Administration. The survey starts in 2010 and has approximately 1000 panelists (households) each year from metropolitan Seoul area, which consists of three provinces (Seoul, Incheon, Gyeonggi-do), the fourth largest metropolitan area in the world with population over 25 million. Panelists receive a mailed journal and are required to record their monthly grocery expenditures. I limit the sample to the balance panel of 639 households that have a non-missing shopping record at the quarterly level from 2010 to 2017. Further, I exclude liquid items, which account for 9% of the total weight of purchased food, that can be discarded down the drain without incurring the tax. I determine the unit tax status for each household using street address information. Having a detailed address is particularly useful because unlike non-condominium residents, whose unit tax status vary at the municipality level, the unit tax status for condominium residents for a typical municipality vary at the complex level. The data documents each purchase in great detail with variables such as type of store, shopping date and time, food category, expenditure, and unit price.¹⁶

Food intake and nutrition. I use the Korea National Health and Nutrition Examination Survey (KNHANES), a nationally representative cross-section survey of approximately 10,000 individuals per year, from the Korea Centers for Disease Control and Prevention. I use responses from metropolitan Seoul area over 2010-2017 that coincides with the grocery purchase data. The data documents food intake—both at and away from home—based on a 24-hour dietary recall face-to-face interview. Further, it converts food consumption data into nutritional intake estimates based on validated scientific methodologies. I supplement the intake and nutrition data with health examination results. Because KHANES dataset discloses address at the community level, which is the smallest administrative unit in South Korea, I can assign the unit tax status at the household level with high precision.

¹⁴The food waste data starts in 2009 and is replaced by another metric in 2016.

¹⁵For municipalities using the smart card system, where the treatment timing could vary at the apartment complex level, I acquire a complex level implementation date through the Official Information Disclosure Act.

¹⁶For shopping records with missing unit price data, I impute the missing values using price information from nearby stores. See Appendix A.2 for more details on imputation steps and validation tests.

Time use on meal production. I use the Korean Time Use Survey from Statistics Korea, an official national statistical agency. The data documents how much time individuals (age over 10) spend on each time-use category in a given day.¹⁷ Time spent on food production is documented in four different categories: cooking, cleaning up after meal, bookkeeping, and shopping. The survey has been conducted once every five years since 1999 and the number of respondents in each survey is about 30,000. It collects two consecutive days of 24-hour time diaries along with data on demographic information. Unlike the grocery or food intake data, time use data discloses respondents’ address at the province—the largest sub-national administrative unit—level, which makes it challenging to assign the unit tax status at the household level. In Section 5, I discuss how I address this limitation.

Municipality characteristics. I use Census on establishments, regional statistics (from the Statistics Korea) and Statistics Yearbook from local governments to collect various municipality characteristics. These variables, which are closely related with waste generating behavior, include number of restaurants, condominium, and non-condominium households, education level, and fraction of single-person household for each year and municipality.

Summary statistics. In Table 3.1, I provide summary statistics for key dependent variables across all observations in the analysis sample. To contextualize different variables, I express everything in terms of per year per household. Two points are worth highlighting. First, comparing the amount of total food waste with the purchased food amount suggests that 28% of the purchased food is discarded, which is consistent with the global average (FAO 2013). Further, the consumed (526kg) and wasted (206kg) food quantity adds up to the purchased quantity (740kg), indicating that the three datasets together successfully capture an average household’s food usage pattern.

Second, the farm-to-kitchen GHG emissions from an average food basket (740 kg) are 2,678kg CO₂eq, which is as large as the emissions from driving 6,849 miles with an average passenger vehicle.¹⁸ To convert food purchases to GHG emissions, I leverage food-item specific farm-to-kitchen GHG emissions data from Poore and Nemecek (2018). Also, 2,678kg CO₂eq suggests that the external cost of food waste due to GHG emissions alone is \$0.69/kg at \$190 social cost of carbon (EPA 2023). However, the highest observed unit tax rate is less than 20% of \$0.69, which is not enough to

¹⁷The number of time categories are 124, 137, and 144 for 1999, 2004, and 2009 survey, respectively.

¹⁸Based on EPA’s Greenhouse Gas Equivalencies Calculator (accessed Aug 5, 2024 at <https://www.epa.gov/energy/greenhouse-gas-equivalencies-calculator>).

Table 3.1: Descriptive Statistics (Per Household Per Year)

Variables	Min.	Max.	Mean	Std.Dev.	N
Food waste (kg)	34.11	780	206	72.55	441
Total grocery purchase (kg)	20.97	2,213	740	307	2,872
Total grocery expenditure (USD)	119	12,312	3,658	1,534	2,872
Total GHG from grocery (kg CO_2eq)	38.03	9,912	2,678	1,270	2,872
Intake at Home (kg)	0	5,142	526	477	17,264
Intake away from Home (kg)	0	8,764	933	737	17,264
Calorie at home (1000 Kcal)	0	9,320	799	624	17,264
Meal Production Time (Hrs)	469	1,067	726	87.21	672

cover the waste management costs alone.

Appendix A.1 provides descriptive statistics for additional variables. Appendix A.2 conducts various validation tests such as assessing the representativeness of survey respondents. Finally, Appendix A.3 reports balance test results.

4 Effect of the Unit Tax on Food Usage

4.1 Estimation Framework

Binary Treatment Models. I exploit the Wave 2 unit tax expansion to causally identify the effect of the tax on household food usage. The baseline two-way fixed effect model is in equation (1). Importantly, I remove always-treated and non-absorbing households to minimize potential contamination of the control group, building on the insights from the recent difference-in-differences literature (Baker et al. 2021, Goodman-Bacon 2021).

$$\log(Q_{imt}) = \beta Tax_{imt} + \mathbf{X}_{imt}\delta + \lambda_{im} + \omega_t + \epsilon_{imt} \quad (1)$$

Q_{imt} are various grocery purchase outcomes for household i living in municipality m at year t . Tax_{imt} is a dummy that takes 1 if a household is subject to the unit tax. It has an i subscript because as described in Section 2.2, the treatment status could vary within a municipality. $\mathbf{X}_{ihmt}$ represents time varying household characteristics: working status, income, housing type, age, and family size. I also include household by municipality fixed effect λ_{im} , which allows household characteristics to vary by municipality. This accounts for the possibilities that households are likely to move when

there is a life event such as starting a new job or experiencing household composition changes that can be correlated with grocery demand. β reflects the impact of the unit tax on grocery purchases.

The key identifying assumption is that in the absence of the unit tax, grocery purchases of treated and control households have parallel trends. While overall similarities in baseline characteristics (Appendix Figure A.3) support this assumption, one might worry about the difference in the baseline likelihood of living in apartment. That is, the effect of the tax might be attributable to the housing type specific shocks that coincide with the unit tax treatment. I show that including housing type by year instead of year fixed effects produce very similar results. I also allow for treatment heterogeneity and check pre-trend using alternative estimators proposed in the literature (Cengiz et al. 2019, Callaway and Sant’Anna 2021, Sun and Abraham 2021, Borusyak et al. 2022).¹⁹

To understand the impact of the tax on food intake and nutritional content, I estimate a variant of equation (1): $\log(M_{ihcmt}) = \beta T_{hcm} + \delta \mathbf{X}_{ihcmt} + \lambda_{hm} + \omega_t + \epsilon_{ihcmt}$ where subscript h indicates the housing type (condominium vs. other types) and c is community, which is the smallest administrative unit in South Korea, and the other subscripts are identical as equation (1). M_{ihcmt} are the range of outcome variables such as intake, calories, and vitamin quantities as well as health outcomes.

While the source of variation is identical as equation (1), three differences are worth pointing out. First, since food intake can vary by individual even within a household, i indicates an individual rather than a household. And $\mathbf{X}_{ihcmt}$ is individual level determinants of food intake: income, age, sex, working status, whether someone has a child, and family size. Second, as the data is cross-sectional, I add municipality by housing type fixed effect (λ_{hm}) instead of the household by municipality fixed effects (λ_{im}). Third, because I do not observe a street address, a binary tax status is assigned based on the community and housing type.²⁰ In the estimation process, I weight the regression using sample weights.

Continuous Treatment Models. To estimate the impact of the tax on food waste quantity, I estimate a continuous treatment version of equation (1) because the level of observation in the waste

¹⁹More specifically, I estimate $\log(Q_{ihmt}) = \sum_{k=-4}^3 \alpha^k Tax_{ihmt}^k + \mathbf{X}_{ihmt}\delta + \lambda_{im} + \omega_t + \epsilon_{ihmt}$ where Tax_{ihmt}^k takes 1 when a household is under the tax in event year $k = t - d$ where d is the policy change year. Because the sample is unbalanced in event time, coefficients near the endpoint give unequal weight to households that experienced the unit tax early or late in the sample. Thus, I impose endpoint restrictions such that $\alpha^k = \underline{\alpha}$ for $k < -4$ and $\alpha^k = \bar{\alpha}$ for $k > 3$.

²⁰For condominium residents, I assign $T_{hct} = 1$ when the fraction of households under the tax in community-year is over 50% and 0 otherwise, which in principle can create non-trivial measurement errors. However, as Appendix Figure D.4 illustrates, community is small enough such that the distribution is bimodal with two modes at near 0 and 1 for most cases. The tax status for non-condo residents do not vary within a municipality.

data is municipality.

$$\log(W_{mt}) = \beta(\%)Tax_{mt} + \mathbf{X}_{mt}\delta + \theta_m + \tau_t + \epsilon_{mt} \quad (2)$$

In equation (2), W_{mt} denotes per household food waste quantity for municipality m in year t .²¹ The fraction of households subject to the unit tax is $(\%)Tax_{mt} = \sum_k (\%)tax_{mkt} z_{mkt}$, where $(\%)tax_{mkt}$ is the fraction of unit-taxed households in $k \in \{\text{condominium, non-condo, restaurants}\}$. z_{mkt} is share of k in each municipality. $\mathbf{X}_{mt}$ are three municipality specific characteristics: educational attainment, fraction of the single-person household, and fraction of the households living in condominiums. θ_m, τ_t are municipality and year fixed effects, controlling for unobserved time-invariant municipality characteristics and overall time trend.

β is the coefficient of interest, which informs the marginal effect of changes in $(\%)Tax_{mt}$. Since the Wave 2 expansion was mandated by the central government, endogenous selection by municipalities is unlikely. Consistent with this, the scale of the expansion appears to have been driven by external factors. As shown in Appendix Figure D.3, municipalities with a higher baseline proportion of condominium residents experienced greater expansion, likely due to having more capacity for growth. I also check the robustness of β from equation (2) using a stacked difference-in-difference approach (Cengiz et al. 2019), which allows me to fix the control group to “clean” municipalities, namely bottom 10% municipalities in terms of $(\%)Tax_{m,2015}$. In the estimation process, I use municipality population as a weight. Throughout the analysis in Section 4, standard errors are clustered at the municipality level.

4.2 Findings

Effect of the unit tax on food waste and purchases. I first report the effect of the unit tax on food waste quantity. In Panel A column (1), I regress $(\%)Tax_{mt}$ on the log of food waste quantity per household. The point estimate indicates that the policy effect is economically large and statistically significant. In particular, when the fraction of households subject to the unit tax changes from 0 to 100%, food waste per household decreases by 19% ($e^{-0.209} - 1 = -0.188$) or 46kg when evaluated at the average waste quantity in the pre-expansion period for low treatment intensity municipalities (244kg). In Appendix Table D.1, I report the result from the stacked DD approach. The estimated effect is

²¹The municipality waste quantity reflects waste from both households and small restaurants. Details on the per household conversion, see Section 3.

consistent with Table 4.1, although the effect size is slightly attenuated at 14% ($e^{-0.152} - 1 = -0.141$).

The large effect in column (1) is consistent with prior studies, which document reductions in MSW by 15% to 40% following the implementation of a unit tax of comparable sizes (Fullerton and Kinnaman 1996, Carattini et al. 2018, Bueno and Valente 2019).²² Relatedly, Homonoff (2018) report that a 5-cent bag tax at retail stores decreases plastic bag usage by 40%.

Figure 4.1 (a) illustrates results in column (1) using binned regression. The horizontal axis is the change in the treatment intensity over time (i.e., $(\%)Tax_{m,2015} - (\%)Tax_{m,2009}$) and the vertical axis shows the corresponding change in per household food waste. The fitted line suggests that municipalities with larger increase in treatment intensity experienced a larger reduction in waste quantity.

To understand the social benefit of food waste reduction, I explore the source of observed food waste reduction. Column (2) of Table 4.1 Panel A shows that the unit tax reduces food purchase per households by 5.5% or 41kg based on the pre-tax average. Comparing 41kg to the observed reduction in food waste quantity (46kg) implies that purchasing less food in the first place is the primary abatement strategy. The comparison also implies that the upper bound of illegal dumping is 11%. Given the lack of consensus on the extent of illegal dumping induced by waste pricing (Bel 2016), which critically determines the desirability of the policy, the 11% provides a useful benchmark.

Similarly, in column (3), I find a 4.6% reduction in grocery spending from the unit tax, which amounts to \$169 savings on annual grocery bill for an average household. This is consistent with an external survey, which find that 31%, 21%, and 10% of households reduce grocery purchases by 0–5%, 5–10%, over 10% after the unit tax (Ministry of Environment 2012). Importantly, savings on grocery expenditures suggest that the unit tax generates a private financial benefit in addition to reducing negative externalities. Further, as discussed in Section 3, purchase quantity is subject to measurement error due to missing unit price data. Finding a similar effect using expenditure strengthens the credibility of estimates in column (2).

²²The tax rates in these studies, adjusted to 2015 dollars, are 2.1 cents/kg in Fullerton and Kinnaman (1996), 4.4 cents/kg in Carattini et al. (2018), and 12 cents/kg in Bueno and Valente (2019). For studies where prices are expressed in terms of volume, I apply a conversion ratio of 0.75 kg per liter.

Table 4.1: Effect of the Unit Tax on Household Food Usage

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Food Waste and Food Purchases								
(%) Unit Tax	-0.209*** (0.055)							
Unit Tax		-0.057*** (0.019)	-0.047** (0.021)	-0.101*** (0.030)	-0.035 (0.022)	-0.058*** (0.018)	0.001 (0.016)	-0.061** (0.027)
Dependent Variable in Log	Food Waste Per HH	Grocery kg Per HH	Grocery Exp Per HH	Perishable kg Per HH	Storable kg Per HH	Kg per Trip Per HH	N Shopping Per HH	GHGs Per HH
Effect In Level	-46	-41	-169	-29	-15	-0.25	0.2	-158
Municipality FE	Yes	No	No	No	No	No	No	No
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Municipality $\times$ HH FE	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	441	2,872	2,872	2,870	2,872	2,872	2,872	2,872
Panel B: Food Intake, Nutrition, and Health Outcomes								
Unit Tax	20.64 (22.69)	37.54 (24.83)	20.04 (32.25)	-52.98* (26.69)	-59.3** (28.37)	2.782 (11.28)	-1.246** (0.524)	-0.332** (0.141)
Dependent Variables	Intake at Home	Calorie at Home	Vitamin A at Home	Intake out of Home	Intake out of Home (Discretionary)	Intake out of Home (Non-Disc.)	Weight	BMI
Mean of Dependent Variables	526	799	307	933	861	86	53	21
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Income Group FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Housing Type $\times$ Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	15,665	15,665	15,665	15,665	15,447	15,447	15,633	15,586

Note:

This table reports the effect of the unit tax on household food waste generation and food purchases (Panel A) and food intake (Panel B). Panel A shows the estimation results from equation (2) using municipality by year level food waste data and results from equation (1) using household level grocery panel data. Panel B shows the estimation results for a variant of equation (1) using individual level food intake data. I only report coefficients for the unit tax term, but baseline control variables are included. Outcome variables in Panel A are in log scale while outcome variables in Panel B are in the original scale. All standard errors are clustered at the municipality level.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

In columns (4) and (5), I separately estimate the policy effect for perishable (fresh fruit and vegetable) and storable food items. If adjustment in food purchases is driven by the unit tax, the policy effect should be larger for perishable items, which are more likely to become waste when not consumed in time. Indeed, I find that the point estimate is over three times larger (in magnitude) for the perishable items. Columns (6) and (7) indicates that the reduction in the grocery purchases is almost entirely driven by purchasing less food per store visit, perhaps in an effort to refrain from purchasing food that will not be consumed.

In Figure 4.1 (b), I show how grocery purchase quantities change relative to the unit tax implementation timing. The point estimates before the unit tax is at 0, satisfying the parallel trend assumption. Also, the treatment effect is sharp and persistent for the next three years. In Appendix Figure D.5, I repeat the exercise using alternative estimation methods that allow treatment heterogeneity (Cengiz et al. 2019, Callaway and Sant’Anna 2021, Sun and Abraham 2021, Borusyak et al. 2022). Overall, I find similar point estimates and standard errors, which suggests that the effect is robust to estimation approaches once always-treated and non-absorbing observations are removed. In Appendix Table D.2, I add housing type by year fixed effects, allowing for the possibility that households in different housing types may respond differently to national-level shocks, and find that the results are remain very similar to those in Table 4.1.

Effect of the unit tax on food intake. How does change in food purchases affect food intake? In Table 4.1 Panel B, column (1), I find that the unit tax leads to a statistically insignificant 3.8% increase in the quantity of food consumed at home. Consistent with this, columns (2) and (3) show slight increases in key nutrient intake at home following the tax. These estimates implies that the reduction in grocery purchases likely comes from the previously wasted portion of the food basket rather than from the consumed portion.

In contrast, the results in columns (4)-(6) show that individuals reduce their consumption of food away from home, particularly from discretionary sources such as restaurants and takeout. This suggests that after the unit tax, individuals substitute meals away from home with meals prepared at home. This behavior may be driven by the tax encouraging closer monitoring and more timely consumption of groceries, which could be at odds with unplanned food purchases or dining out. Although the food intake data do not include information on the monetary expenditure on food items,

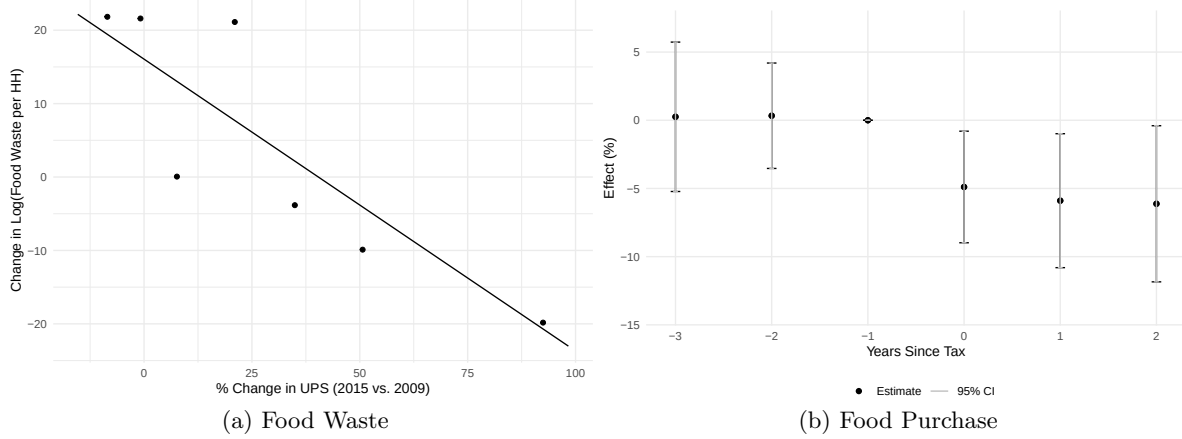


Figure 4.1: The Effect of the Unit Tax on Food Waste and Purchases. Panel (a) shows results from a binned regression estimation between the change in the fraction of households under the unit tax between 2009 and 2015 and the change in the food waste quantity per household. Panel (b) shows an event study plot for food purchase quantity. All dependent variables are log transformed.

the reduction in food consumed away from home indicates an additional financial benefit of the unit tax, beyond the savings on grocery expenditures. Finally, columns (7) and (8) indicate that changes in food intake result in a modest reduction in weight and BMI. This effect may be driven by a decrease in food consumed away from home, which is often associated with poorer dietary quality, including higher levels of salt and saturated fats (Gesteiro et al. 2022).

5 Household Abatement Strategies and Corresponding Costs

5.1 Estimation Framework

The findings in Section 4.2 suggest that the unit tax encourages a more efficient food use without compromising nutritional needs. But how households can maintain food intake with smaller grocery purchases? What do they do and what is the associated costs? To answer these questions, I build on the insights from the household production model (Becker 1965).

$$M = AF(Q, T) \quad (3)$$

Consider a simple meal production process in equation (3), where meal at home (M) is produced by combining raw food input (Q), namely grocery, and time (T). This household aims to minimize the cost of production by choosing an optimal input mix. Suppose that before the tax, household was

producing M_0 by combining Q_0 and T_0 . Suppose that the corresponding productivity was A_0 . Further posit that after the tax τ is imposed on food waste, household produces at $M_1 = A_1 F(Q_1, T_1)$.

From Section 4.2, I find that $Q_0 > Q_1 = 0.95Q_0$ while $M_1 \geq M_0$, and the model provides two potential explanations for this empirical finding. First, household use more time to compensate for reduced grocery purchases ($T_1 > T_0$). Second, the unit tax might have increased the productivity, allowing households produce the same amount of meal using smaller amount of grocery ($A_1 > A_0$).

To estimate the impact of the unit tax on time spent on meal production, I analyze microdata from the 1999, 2004, and 2009 Korean Time Use Surveys. Unlike grocery or food intake data, which provide detailed location information for respondents, the time use data report locations only at the province level—the largest sub-national administrative unit in South Korea. Consequently, the best available measure of individual households' tax status is the proportion of households subject to the tax within each province-housing type.

Importantly, because almost all province-housing types experienced substantial unit tax expansions during the Wave 2 period (Appendix Figure D.6 Panel (b)), due to the nature of the central government's mandate, identifying a clean control group during this period is particularly challenging.²³ To address this, I leverage the Wave 1 expansion, where most province-housing type pairs remained untreated (Appendix Figure D.6 Panel (a)). Further, I collapse individual-level data into province-housing type-survey date-year level observations.²⁴

$$Time_{hpd} = \beta(\%)T_{hp,2009}I_t + \delta\mathbf{X}_{hpt} + \alpha_{hp} + \lambda_t + \omega_d + \epsilon_{hpd} \quad (4)$$

A resulting canonical difference-in-differences estimating equation with continuous treatment is in equation (4). $Time_{hpd}$ is time spent on meal production activities per household for housing type by province hp on day of the week d in year t . $(\%)T_{hp,2009}$ is the proportion of households subject to the unit tax as of 2009, or the time-invariant treatment intensity, for hp . I_t is a post period indicator that takes 1 if t is 2009, and $\mathbf{X}_{hpt}$ represents the determinants of household time allocation, including factors such as whether the household has children, family size, employment status, gender, and house size, which serves as a proxy for income. In the estimation process, I weight the regression

²³Note, identifying a control group is not a concern in Section 4, as the granular location information allows for the assignment of a binary tax status to each household following the Wave 2 expansion.

²⁴I show that analysis without data aggregation produces similar results.

by the number of households in *hpt*. Standard errors are clustered at the province by housing type. Details on the estimation approach and results for the meal production productivity effect are in Appendix B.2.

5.2 Findings

In Table 5.1, I estimate the impact of the unit tax on the time spent on meal production activities. In column (1), I show that an average household spend 9.1 (7.6%) additional minutes per day or 55 additional hours per year on meal production after the unit tax. Investigating individual activities in columns (2)-(5), I find that the result in column (1) is driven by higher meal preparation time, which include cooking, preparing ingredients, storing and organizing groceries, and setting the table (Statistics Korea 2009). I also find a statistically significant increase in time spent on keeping diaries, which facilitates meal planning. There is evidence of increased time spent on cleaning as well, but the effect fails to rule out the null effect at the conventional statistical significance level.

Interestingly, column (5) indicates that time spent on grocery shopping does not increase, which is consistent with findings from Section 4.2 that the tax does not increase the number of shopping trips. This suggests that higher food use efficiency does not come at the cost of increased GHG emissions from more trips (i.e., more driving). Taken together, households seem to use more time to compensate for fewer groceries, and such a substitution between time and money (although by different triggers) has been well documented in earlier studies (Aguiar and Hurst 2005, 2007a). Appendix Table D.3 shows that the estimated results from uncollapsed data lead to the identical conclusion.²⁵

Figure 5.1 (a) visually illustrates Table 5.1 column (1). For this, I run binned regression between $(\%)T_{hp,2009}$ and the first difference in meal production time between 2004 and 1999 (pre period) and 2009 and 2004 (post period). Additionally, the difference for the bin with the lowest treatment intensity is normalized to zero to enhance readability. The left-hand side figure shows that the change in time spent on meal production between the high versus low treatment intensity units are essentially the same, which is akin to the “no pre-trend” condition for a binary canonical difference-in-differences case. Importantly, the post period (2004 vs. 2009) shows a positive slope, which suggests that households in the high treatment intensity units are spending more time in meal production

²⁵The magnitudes are slightly attenuated perhaps due to measurement errors in the binary tax status variable, which is defined as 1 if the fraction of households under the unit tax in a province-housing type is over 50%.

Table 5.1: The Effect of the Unit Tax on Household Meal Production Time

	(1)	(2)	(3)	(4)	(5)	(6)
(%) Unit Tax x Post	9.07** (4.03)	7.49*** (2.22)	2.31 (1.39)	0.438** (0.192)	-1.16 (1.37)	0.479** (0.224)
Dependent Variable	Overall Time	Prep Time	Cleanup Time	Diary Time	Shopping Time	Dollars Per Day
Province $\times$ Housing Type FE	Yes	Yes	Yes	Yes	Yes	Yes
Survey Date FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	672	672	672	672	672	448

Note:

This table reports the effect of the unit tax on household meal production time by estimating equation (4) using the time use survey of 1999, 2004, and 2009 (2004 and 2009 for column (6)). I report the coefficient of interest only, but all the regressions include baseline control variables. Standard errors are clustered at the province by housing type level. $*p < 0.1$; $**p < 0.05$; $***p < 0.01$.

than their low treatment intensity counterparts after the unit tax.

To explore how households utilize the additional time allocated on meal production, I leverage web search data from “Naver,” a leading search engine in South Korea.²⁶ In Figure 5.1 (b), I depict web search intensity for three keywords: “food waste,” “organizing refrigerator,” and “meal planning,” for female users aged 30–60, a demographic primarily responsible for meal production. The figure shows that search intensities for “organizing refrigerator” and “food waste” are strongly correlated over 2017–22. Further, the search intensity for “organizing refrigerator” sharply increased during COVID-19 lock down in 2020, when individuals had to prepare more meals at home. Interestingly, as Appendix Figure D.7 illustrates, such a pattern does not appear at all for other demographic groups (men aged 30–60 or female younger than 20), which collectively suggest that organizing the refrigerator is one of the key strategies for reducing food waste and improving food use efficiency.²⁷ This aligns with earlier findings that proper food storage to maximize shelf life is the most common way consumers try to reduce wasted food (IFIC Foundation 2019).

In Table 5.1 column (6), I estimate the abatement cost in dollars by regressing equation (4) using

²⁶If households had perfect knowledge of food waste reduction strategies, the unit tax would simply motivate them to apply what they already know, making search data less informative for this purpose. However, survey results reveal a knowledge gap in these strategies (Ministry of Environment 2015b).

²⁷Because search intensities are normalized for each demographic, I cannot compare them across different groups.

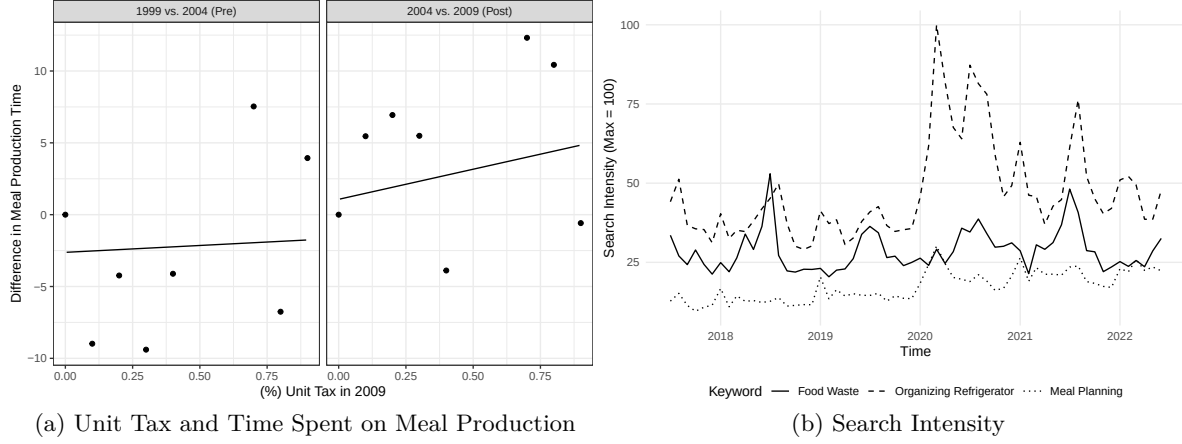


Figure 5.1: The Effect of the Unit Tax on Time Use. Panel (a) shows the impact of unit tax on time spent on meal production using binned regression. The X-axis shows the fraction of households under the unit tax in 2009 and the Y-axis shows the first difference in meal production time for years 2004 vs. 1999 (left) and 2009 vs. 2004 (right). Panel (b) shows the trend in web search intensity for three food waste related keywords (food waste, organizing refrigerator, and meal planning) for female users aged 30-60 between July 2017 and Jun 2022 from Naver, a dominant search engine in South Korea.

the value of time spent on meal production as an outcome variable. To measure the opportunity cost of time, I follow Aguiar and Hurst (2007a) and use demographic group specific marginal returns to shopping (see Appendix B.1 for more details). I find that the cost of additional time spent on meal production is \$175 per year per household, which almost exactly offset by savings on grocery spending (\$169 from Table 4.1) due to the unit tax.²⁸

Finally, I present suggestive evidence indicating that the unit tax does not affect meal production productivity. While I leave details to Appendix B.2, I highlight that the null effect on productivity is not surprising given that productivity is measured by the residual in the production function. That is, when a 5.5% reduction in grocery purchases is compensated by a 7.6% increase in time use (while output level does not decrease), a large increase in productivity seems implausible.

6 Discussion

6.1 Welfare Effects

The empirical estimates allow me to conduct a simple cost-benefit analysis of the unit tax policy.

For this, I assume that the food production market is competitive, and thus the social welfare of the

²⁸\$175 suggests that the average value of time in the context of home production is \$2.9 per hour. As a benchmark, the average market wage in South Korea for homemaking jobs in 2009 was \$4.9 per hour.

tax is determined by its impact on GHG emissions, government spending, and consumer surplus. I abstract away from the tax revenue because it is a transfer between households and government.

Reduction in GHG emissions.: Using estimates from Table 4.1, Table 6.1 shows that the unit tax reduces lifecycle GHG emissions from wasted food by 3.1 million tons annually, which equivalent to the emissions from 750 thousand passenger vehicles (for more details, see Appendix C). Importantly, this effect is almost entirely driven by avoided production (item A: 3.0 million ton), which highlights the importance of food waste prevention over recycling.²⁹ Note, this exercise relies on two assumptions. First, profit-maximizing farmers will scale back food production in response to the unit-tax induced demand shock. Second, any food waste reduction not explained by lower grocery purchases will end up in landfills, which is the worst-case scenario in terms of GHG emissions.

Government spending.: Since food waste tax revenue covers less than 50% of food waste pickup and treatment service costs, local governments spend a large portion of their budget to provide these services—for instance, the Seoul Metropolitan Government spent \$136 million in 2015 alone. Assuming a constant marginal cost, a 19% reduction in food waste after the unit tax saves \$79 million per year nationwide even after accounting for increased landfill costs under the worst case leakage scenario. For a complete impact on government budget, the unit tax implementation costs need to be considered as well. In Table 6.1, I take into account both the operating (item D and F in Table 6.1) and capital cost (item E in Table 6.1) of the unit tax system.³⁰ The combined program cost (items D+E+F) amounts to \$36 million per year. Note, I exclude spending on waste pickup and treatment services, which must be provided regardless of the tax policy. Taken together, the unit tax saves government spending on food waste by \$43 million per year.

Abatement costs.: The primary abatement cost of the unit tax is the effort exerted by households. However, Table 4.1 implies that savings on grocery spending offset the effort costs. To systematically evaluate the overall abatement costs, I consider a utility function $U = X + \delta L + v(M)$ from a household

²⁹The GHG reduction effect from waste treatment can be much larger in other countries. In South Korea, nearly all food waste is processed at specialized facilities, which generate significantly less GHG than food waste in landfills. However, this practice is an exception rather than the norm; for instance, in the U.S., 50% of total food waste ended up in landfills in 2018 (EPA 2020).

³⁰For items D and G in Table 6.1, I use the 2015 Unit-Based Waste Policy Yearbook, which is the first year with detailed cost breakdown. For E and F, I use smart card system installation data from Korean Environment Corporation (<https://www.citywaste.or.kr/EgovPageLink.do?link=/ucwmsNew/portal/sysInfo/sysInfo06>, accessed on Dec 5, 2023).

Table 6.1: Welfare Effect of the Unit Based Food Waste Tax

	Items	Value
GHG Change (1000 Ton)	Avoided Production (A)	3020
	Avoided Food Waste Treatment (B)	167
	Increased Food Waste in Landfill (C)	63
	Net GHG Effect (A+B-C)	3123
Government Spending (\$ Mn)	Producing Bags and Stickers (D)	12
	Installing Smart card System (E)	17
	Operating Smart card System (F)	7
	Spending on Waste Services (G)	-79
	Total Gov. Spending (D+E+F+G)	-43
Household Abatement Costs (\$ Mn)	Change in Consumer Surplus (H)	115
Total Abatement Costs/Ton of CO₂eq	(D+E+F+G+H)/(A+B-C)	23
Net Benefit (\$ Mn)	(A+B-C) × \$190 - (D+E+F+G+H)	522

production model (Becker 1965). Here, the total utility U is sum of numeraire X , utility from leisure δL (δ is the marginal value of time), and utility from meal quantity $v(M)$, where M is produced by combining time input T and raw food input Q (i.e., $M = F(T, Q)$). The quasilinear functional form reflects the fact that money spent on groceries and time spent on meal production is relatively small portion of the total budget and time endowment (see as Table 3.1). A consumer is subject to two constraints: $X + PQ = I$ where P is price of grocery, I is total income, and $T + L = \bar{T}$ where $\bar{T}$ is total non-working time. By plugging in two budget constraints into the utility function, household's problem reduces to $U = v(M) + I - PQ + \delta(\bar{T} - T)$.

Posit that before the unit tax, households chose Q_0 and T_0 to attain the utility level of $U_0 = v(M_0) + I - PQ_0 + \delta(\bar{T} - T_0)$. Now suppose that the unit tax is imposed and households choose Q_1 and T_1 to attain the utility level of U_1 . Then $U_1 - U_0 = v(M_1) - v(M_0) - P(Q_1 - Q_0) - \delta(T_1 - T_0)$. Using estimates from Sections 4 and 5, I know that $v(M_1) \geq v(M_0)$, change in grocery spending $P(Q_1 - Q_0) = -169$, and change in time costs $\delta(T_1 - T_0) = 175$, which implies that $U_1 - U_0 \geq -6$ (i.e., the maximum annual utility loss is \$6 per household) assuming that $v()$ is a monotonically increasing in M . Thus, the maximum household abatement cost at the national level is \$115 million (Item H).

Overall impacts.: Table 6.1 provides two key metrics to assess the welfare effect of the unit tax. First, the abatement cost per ton of CO_2eq is \$23, which is calculated by dividing the total GHG

reduction by the total economic cost (government spending and household abatement costs) of the policy. Notably, \$23 per ton indicates that the unit tax is one of the most cost-effective GHG mitigation measures (Gillingham and Stock 2018). Second, the net benefit of the policy is \$522 million per year. For benefit, I monetize the benefit of reduced GHG emissions using a social cost of carbon of \$190 per ton of CO_2eq (EPA 2023). From this, the household abatement cost of \$115 million is subtracted. Importantly, the \$522 million likely underestimates the true benefits of the policy, as it does not account for additional environmental benefits of food waste prevention, such as reduced biodiversity loss, soil degradation, and water depletion (EPA 2021).

6.2 Designing Food Waste Taxes

While lifecycle GHG emissions vary substantially across food items—for example, beef is 100 times more carbon intensive than vegetable (Poore and Nemecek 2018)—the unit tax imposes a uniform tax rate for every wasted food. Also, the costs of measuring individual emissions (i.e., food waste quantity) may deter other governments to introduce a tax on food waste. Given these limitations, a tax on a market good that is related to emissions, namely a carbon tax on food can be considered as an alternative policy (Fullerton et al. 2010). Despite its theoretical possibility, however, implementing such a tax in practice is difficult for at least three reasons.

First, unlike CO_2 where emissions can be measured based on fossil fuel usage (due to emissions are proportional to fuel consumption), there is no straightforward method to measure non- CO_2 GHG emissions from food production (Timilsina 2022). For instance, emissions from growing beef or rice depends heavily on agricultural practices, which can substantially vary even at the individual farm level, are extremely costly to measure. Second, taxing food consumption itself rather than penalizing wasting food is likely to be politically contentious with concerns over food insecurity and regressivity (Godfray et al. 2010, Dechezleprêtre et al. 2022).³¹ Third, tax saliency literature suggest that the food tax, which is likely to be reflected in the final price consumers pay, might not have as strong effect as a food waste tax (Finkelstein 2009, Chetty et al. 2009). Given these considerations, a unit based tax on food waste is likely to be more realistic and effective policy apparatus than taxing food.

³¹Indeed, the federal government of Canada exempted its agricultural sector from a national carbon tax (Wu and Thomassin 2018).

7 Conclusion

The lifecycle GHG emissions from wasted food are comparable to those of road transport (IPCC 2014, FAO 2015). Additionally, waste treatment is the largest budget item for many municipalities worldwide, often diverting resources from more productive investments (OECD 2000, Kaza et al. 2018). As a result, addressing food waste has become a critical priority for both climate change mitigation and public finance management. While imposing a corrective tax on food waste generation is a textbook solution, limited evidence exists on its welfare effects.

By leveraging two waves of plausibly exogenous food waste tax expansions in South Korea, I show that the tax reduces annual food waste and grocery purchases by 19% (46kg) and 5.5% (41kg), respectively. This enhanced efficiency in food use leads to an annual reduction of 3.1 million tons of CO₂eq (GHG emissions) and \$43 million in government spending. However, these efficiency gains come with time costs: after the tax, households spend more time on meal production, amounting to an annual cost of \$175 per household. Importantly, these time costs are largely offset by savings on grocery spending (\$169), which substantially moderates the overall abatement cost. The findings of this paper show that the unit tax is a highly effective in reducing both externalities at a moderate costs, and thus greatly enhances social welfare.

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A Data Appendix

A.1 Detailed Descriptive Statistics

Table A.1: Descriptive Statistics for Food and Time Use

Variables	Min.	Max.	Mean	Std.Dev.	N
Panel A: Food Waste					
Number of residential households	4,993	420,976	128,226	76,591	441
Number of restaurants (smaller than 2,152 ft^2)	315	13,899	4,775	2,613	441
Number of combined households (HH)	7,198	518,269	161,655	93,299	441
Food waste (kg)	34.11	780	206	72.55	441
Landfill waste (kg)	36.12	878	263	101	441
Panel B: Food Purchases					
Total grocery purchase (kg)	20.97	2,213	740	307	2,872
Perishable grocery purchase (kg)	0	1,363	296	173	2,872
Storable grocery purchase (kg)	15.55	1,553	443	187	2,872
Total grocery expenditure (USD)	119	12,312	3,658	1,534	2,872
Number of trips	10	354	171	54.43	2,872
Total GHG from grocery (kg CO_2eq)	38.03	9,912	2,678	1,270	2,872
Panel C: Food Intake and Nutrition					
Intake (kg)	34.45	8,764	1,460	752	17,264
Intake at Home (kg)	0	5,142	526	477	17,264
Intake away from Home (kg)	0	8,764	933	737	17,264
Discretionary Intake (kg)	0	8,764	861	723	17,009
Non-discretionary Intake (kg)	0	2,705	86.06	208	17,009
Calorie at home (1000 Kcal)	0	9,320	799	624	17,264
Calorie away from Home (1000 Kcal)	0	11,156	1,097	860	17,264
Vitamin A (g)	0	26,164	307	520	17,264
Weight (kg)	7.92	128	53.33	18.16	16,240
Panel D: Meal Production Time					
BMI	11.39	40.96	21.35	4.02	16,192
Meal Production Time (Hrs)	469	1,067	726	87.21	672
Cooking Time (Hrs)	243	562	403	47.26	672
Cleaing Time (Hrs)	98.85	304	216	29.06	672
Diary Time (Hrs)	0	60.83	6.17	5.86	672
Shopping Time (Hrs)	14.36	276	101	40.23	672
Non-food Home Making Time (Hrs)	365	1,217	653	97.43	672

Table A.1 presents summary statistics for a full set of variables on the food and time usage. The variables are grouped into four different categories: wasted, purchased, consumed food, and time use. Each panel merits discussion. For Panel A, the waste quantity in the Unit-Based Waste Yearbook data reflects waste from both residential households and small restaurants. To calculate per household food waste quantity, I treat a restaurant as seven households based on statistics from the City of Seoul. The first three rows present summary statistics for the number of residential households, restaurants, and both (residential and restaurant-converted households). The fourth and fifth rows

jointly show that food waste accounts for 44% of the overall (food and landfill) waste quantity, which is a general pattern found in many countries.³²

The second panel presents descriptive statistics for grocery purchases. An average panelist purchases 740kg of groceries per year, spending \$3,658. To translate expenditure to quantity, I divide expenditure on each food item by its unit price. When compared against the amount of food waste generated, it means that 28% of the purchased food is discarded. This is consistent with findings from the FAO that 1/3 of the produced food is wasted globally (FAO 2013). When I split up food categories into perishable (fresh vegetable and fruits) and storable items, Table A.1 shows that 40% of the total purchase is perishable items. To make these purchases, households make a grocery trip every two days (or 171 trips per year).

The third panel presents descriptive statistics on food intake, nutrition, and body metrics. To make the comparison with Panel A and B easier, I converted per capita daily intake quantities to per household annual intake quantities.³³ The first three rows in Panel C indicate that an average household intakes 1,460kg of food and about 40% of them are consumed at home. Importantly, the total intake quantity is consistent with statistics from a government agency.³⁴ The next two rows show that only about 10% of food consumed away from home is non-discretionary, such as meals at school or office cafeterias, suggesting that there is potentially a large room for food usage adjustments in response to the unit tax. The next row shows that an average household acquires 799,000 Kcal of calories per year (equivalently 826 Kcal per day per person) from food consumed at home.

Finally, Panel D documents annual time spent (in hours) on meal production for an average household—the total amount of time spent by any adult (age over 19). The first row shows that an average household spends two hours per day on meal production. From the second to fifth rows, I split up meal production into smaller time categories, and find that over 85% of the meal production time is time spent inside of kitchen—preparing ingredients, storing groceries, cooking, setting table, and doing the dishes (Statistics Korea 2009). Households spend another 100 minutes per day on other home making activities, including cleaning, laundry, and organizing/sorting. These numbers are consistent with other studies that have used the time use survey to document trends in nonmarket working hours in Korea (Seo et al. 2021). Further, the number of hours spent on home making is comparable to that of the US. For instance, Aguiar and Hurst (2007b) finds that in 2003, an individual in the US spent 118 minutes per day on various home making activities.³⁵

A.2 Data Validation

Assessing the representativeness. The empirical exercises in this study draw on three distinct household-level datasets. To evaluate the representativeness of the survey respondents, I compare key demographic characteristics with benchmark data, such as the Census or other national-level surveys. Specifically, for age, family size, and housing type (if residing in an apartment) I use the Population and Housing Census, respectively. For employment and income, I use nationally representative survey data “Economically Active Population Survey” and “Household Income and Expenditure Survey (HIES)”, respectively.

³²Kaza et al. (2018) find that food and green waste is 32-56% of the total waste. In general, the proportion is higher for lower income countries.

³³An average household size in the sample is 2.65.

³⁴See <https://www.khidi.or.kr/kps/dhraStat/result5?menuId=MENU01657&gubun=age1&year=2017> accessed on Aug 5, 2024.

³⁵These activities include “core” nonmarket work, which consists of meal production, laundry, indoor household cleaning, and “obtaining goods and services”, which include grocery shopping, shopping for other household items, running errands, and buying goods online. They also include corresponding travel times in calculation.

Table A.2: Sample Moments vs. Benchmarks

Variables	Mean	SD	Benchmark
Panel A: Grocery Panel			
Age	46.97	8.37	46.14
Housing Type (if APT)	0.54	0.5	0.5
Family Size	3.52	1.18	3.05
If Working	0.55	0.5	0.5
Monthly Income (USD)	3633	1782	3563
Panel B: Food Intake			
Age	39.79	22.95	38.14
Housing Type (if APT)	0.5	0.5	0.5
Family Size	2.91	1.26	2.65
If Working	0.59	0.49	0.6
Monthly Income (USD)	3105	2196	3217
Panel C: Time Use			
Age	35.81		35
Housing Type (if APT)	0.4	0.49	0.45
Family Size	2.69	1.14	2.95
If Working	0.61	0.49	0.58
Monthly Income (USD)	2515	2581	2536

Appendix Table A.2 Panel A presents this comparison for the grocery purchase dataset. To ensure a meaningful benchmark, I restrict the benchmark data to match the demographic profile of the grocery panel, namely (1) 97% of the panelists are female, and (2) the age range is limited to 25–72. The results in Panel A indicate that the grocery panel data largely align with the benchmark. For example, the average age in the grocery panel is 47, closely mirroring the Population Census average of 46.1. The most notable discrepancy arises in family size: the grocery panel reports an average family size of 3.52, while the Population Census indicates an average of 3.05. This difference likely stems from the fact that the grocery panel did not survey single-person households until 2017, the final year of my sample. In empirical exercises, I control for family size to minimize the potential bias from different family sizes.

In Panel C, I compare the time use data to benchmarks from the 1999–2009 period. Again, sample means closely align with benchmarks, suggesting that the time use data successfully represents an average household’s time usage pattern.³⁶

Additional validation tests for the grocery panel data. There are two potential concerns with the grocery panel data. First, the data might capture only a subset of the panelists’ shopping behavior. This can happen when households fail to keep the record of every single spending. Although the Rural Development Agency compensates panelists \$50 per month and replaces unreliable households,

³⁶The time use survey provides age data for individuals aged 10 and older, while for those younger than 10, it records only the number of preschoolers in each household. To estimate the mean age of the entire population, I assume an average age of 3 years for preschoolers. Although this assumption enables the calculation of the overall average age, determining the standard deviation requires further assumptions about the variance in the ages of preschoolers. This limitation has minimal impact to the conclusion since the comparison of sample means indicates that the time use survey already closely mirrors the Population Census.

it could still be the case that households forget or skip reporting. Second, as discussed in section 3, I impute unit price information for shopping records with missing information. In this section, I investigate the validity of the consumer panel data from the two aspects.

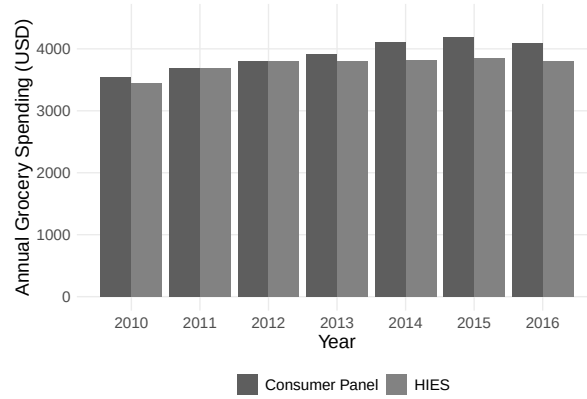


Figure A.1: Consumer Panel Data Validation (Total Expenditure). This figure compares the overall household spending from the consumer panel data and the Household Income and Expenditure Survey.

For the first issue, I compare overall spending amount (in dollars) from an average panelist to household spending information from HIES, a nationally representative survey of over 7,000 households. I use total grocery expenditure net of liquid categories (cooking oil and dairy) from urban households with family size larger or equal to two to make it comparable to the consumer panel. Figure A.1 shows that annual grocery expenditure from these two different surveys are very close to each other over the 2010–16 period.³⁷

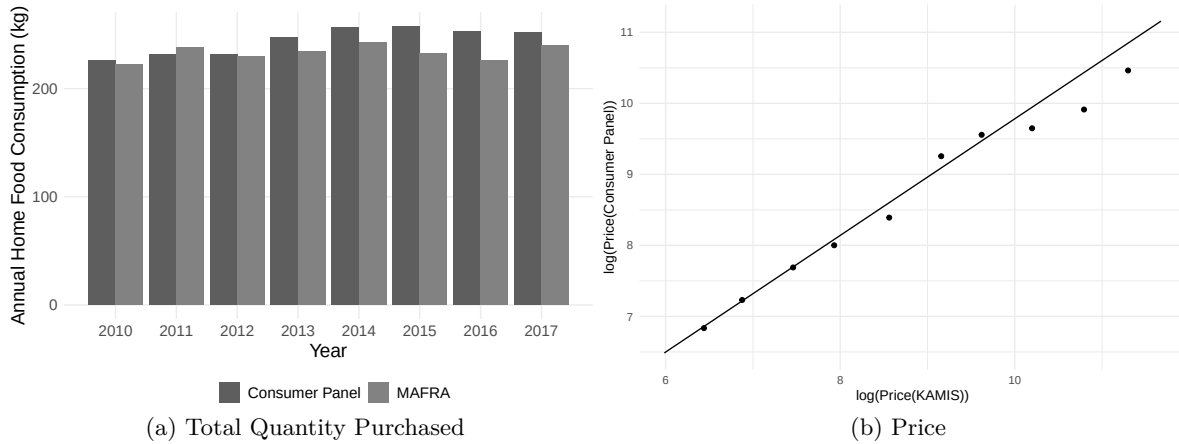


Figure A.2: Consumer Panel Data Validation (Purchase Quantity and Unit Price). These figures examine the validity of the consumer panel data based on total quantity purchased and grocery price.

To address the second issue, namely missing unit price information issue, I conduct two rounds of imputations using a similar approach to Golan et al. (2001). Namely, I first use the median unit price of the same food category from the same type of stores (e.g., farmer’s market, supermarket chains, and mom and pop stores) located in the same municipality and month. This successfully

³⁷2017 is excluded because the grocery panel sample has changed to include a single-person household. The balanced grocery panel does have a very small number of single-person households so I did not use 2017.

imputes 64% of the missing price information. For the second round, I expand the geographic scope to a cluster of 5-6 municipalities and repeat the same exercise. This recovers another 17% of the missing price information. By dividing the total expenditure with the unit price, I back out the quantity purchased.

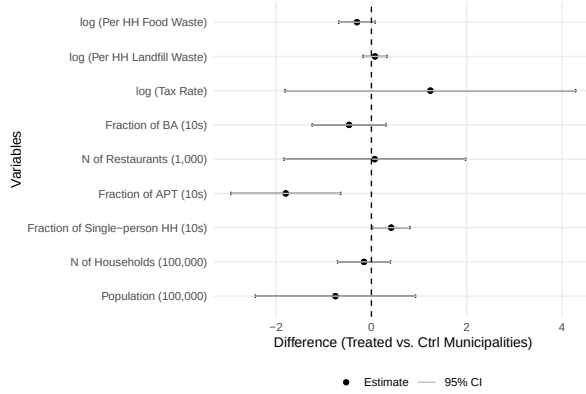
To test the validity of this procedure, I compare the per household grocery purchase in kg from the grocery panel and per household food consumption statistics from the Ministry of Agriculture, Food and Rural Affairs (MAFRA). Importantly, the consumption statistic does not distinguish food that is actually consumed or eaten versus leftover. As the MAFRA data covers food consumption from both home and outside (e.g., restaurants, cafeteria, etc), I adjust it using the fraction of meal consumed within home from Han (2018).

Panel (a) of Figure A.2 shows the result. From 2010 to 2017, the amount of food purchased between the consumer panel and MAFRA official statistic are very closely related. This add credibility to the unit price imputation. In panel (b), I provide binned scatterplot between the unit price information from the grocery panel and the price information from KAMIS (Korea Agricultural Marketing Information Service) website, which is an official source, maintained by the Korea Agro-Fisheries & Food Trade Corporation. The figure suggests that the imputed price is highly correlated with the actual price.

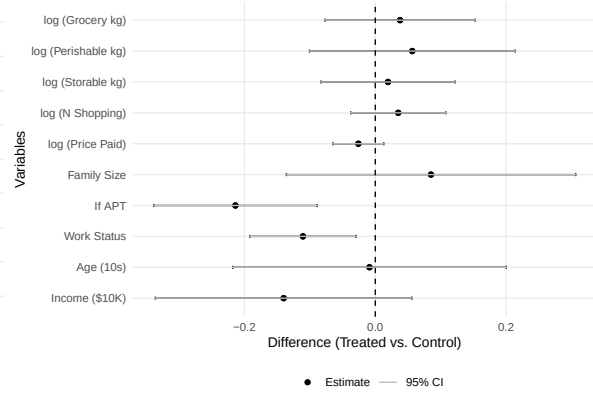
A.3 Balance Tests

Balance plot. In Appendix Figure A.3, I show various balance test results to investigate if there are baseline differences between the treated and control observations. In Panel (a), I compare differences at the municipality level using the waste data. Here, the control group consists of bottom 10% or 7 municipalities in terms of the % of households subject to the unit tax in 2015, while the treated group has the rest of 56 municipalities. I show that the treated and control municipalities are highly comparable in various observed characteristics in 2009, which is the first year in the waste quantity sample. For instance, food and landfill waste quantities are not statistically significantly different from each other. Further, numerous demographic variables show similarities. One important exception is the fraction of apartment residents. It is substantially lower for the treated municipalities, and this is not surprising given that many non-apartment residents were required to use trash bags during the Wave 1 expansion. In panels (b)–(d), I compare ever versus never treated households or individuals for various food and time usage and demographic variables in the pre-treatment period. Consistent with panel (a), I find that observed characteristics in general are well balanced except for the housing type (i.e., fraction of apartment residents). While the baseline characteristics seem similar between the two groups, one might still worry that households might differ in unobserved characteristics by housing type. To account for this, I include housing type specific fixed effects in empirical analyses except.³⁸

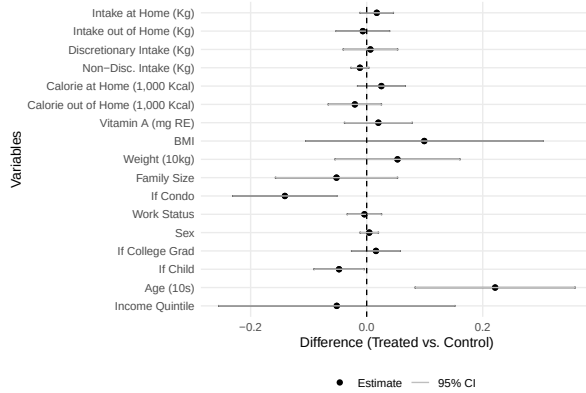
³⁸I cannot implement this with food waste because the data is observed at the entire municipality level.



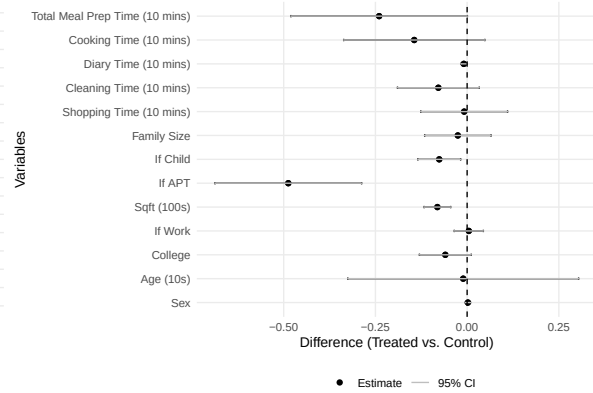
(a) Balance: Food Waste



(b) Balance: Grocery



(c) Balance: Intake



(d) Balance: Time Use

Figure A.3: Balance Plots. These figures compare treated vs. control units on pre-treatment outcome variables and demographic characteristics. Panel (a) shows the difference between treated and control municipalities for observed characteristics in 2009. Control municipalities consists of bottom 10 percent municipalities in terms of the fraction of households subject to the unit tax in 2015. Panels (b)–(d) show the comparison of key variables between ever and never treated households from the grocery panel, food intake, and time use data.

B Details on Household Abatement Strategies

B.1 Calculating the Dollar Value of Increased Time Spent on Meal Production

In Table 5.1 column (6), I estimate the impact of unit tax on household abatement costs. For this, I need to express the outcome variable in equation (4) in dollar value of time spent in meal production. A crucial step in this exercise is to pin down the opportunity cost of time, and I use the *marginal returns to shopping*—how much money households can save by spending more time on searching for cheaper items—as the value of time following Aguiar and Hurst (2007a).³⁹ The choice seems more appropriate than wage, which is frequently used to measure the opportunity cost of time, especially within the household production context—over half of the primary meal preparers in the sample are not formally employed and thus wage is not well defined.⁴⁰

The marginal return on shopping is calculated by multiplying (1) the elasticity of price with respect to shopping frequency and (2) average spending per shopping trip. To express the value of time on an hourly basis, rather than per shopping trip, I adjust the estimated value by multiplying the number of trips per hour.

For the elasticity, I assume that it is identical across different households and take the headline estimates (0.1) from Aguiar and Hurst (2007a).⁴¹

For the average spending per shopping trip, I use information from the grocery shopping data because the time use data do not have information on spending. I start by aggregating the grocery panel data by demographic characteristics—income (low if monthly income is below USD 1,818), family size (1-2, 3-4, and above), and age (20-30, 30-50, and above)—and calculate spending per shopping trip for each demographic cell. Then, I aggregate the time use data using same set of demographic characteristics and merge it with the average spending information to calculate the average number of shopping per hour for each cell. Then, I calculate the value of time for each cell by taking the product of elasticity (0.1), average spending per trip, and average number of trips per hour. Finally, I merge the VOT estimates with the time use data based on demographic characteristics. Note, as the time use data has income variable from 2004 and onward, for this exercise, I limit the analysis to 2004 and 2009 time use survey data.

One potential concern for using the elasticity from Aguiar and Hurst (2007a) is to what extent Korean households are similar to the US households. One of the key findings in Aguiar and Hurst (2007a) is that the price of time substantially varies with age and income. That is, the opportunity cost of time is highest in the middle age, which usually involves disproportionately large responsibilities at work and home. Similarly, the value of time is higher for higher income groups. To test if the Korean consumers exhibit similar characteristics, in Appendix Figure B.1, I create bin scatter plots between (a) age and (b) income decile against the value of time, normalized with the lowest group's value. Consistent with Aguiar and Hurst (2007a), I find that the value of time has an inverse-U shape with age and is positively correlated with income.

³⁹Hastings and Shapiro (2018) and Nevo and Wong (2019) also used returns to shopping to measure the opportunity cost of time.

⁴⁰Aguiar and Hurst (2007a) points out two additional concerns: (1) the observed wage may not be the marginal return to labor (due to nonlinear wage schedules, human capital accumulation on the job, etc.), and (2) individuals may not be able to adjust labor hours freely at the margin.

⁴¹I choose to not to estimate the elasticity because the grocery panel data does not have a UPC code. Without a UPC code, I cannot tell whether the observed relationship between shopping time and paid price is due to price differences for the identical product or product differences between stores (e.g., Coke vs Pepsi).

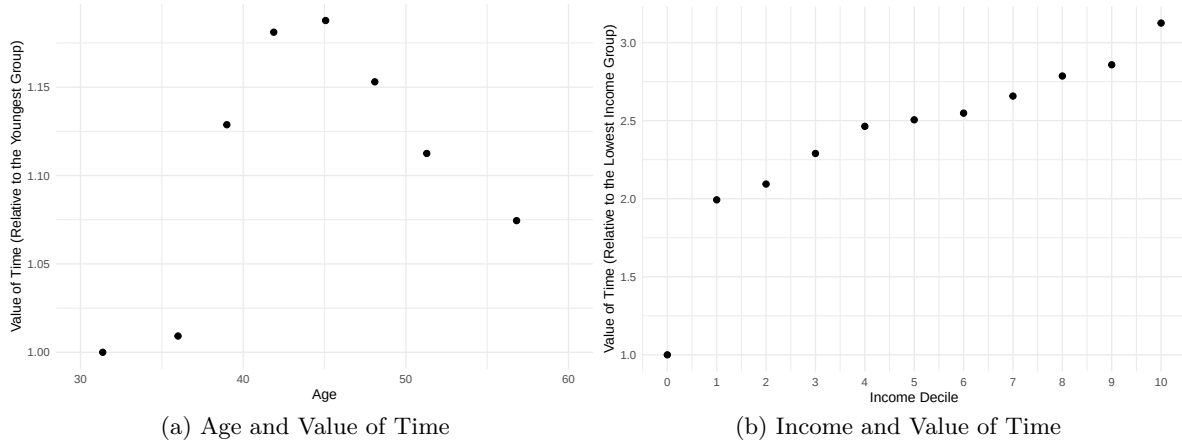


Figure B.1: Value of Time by Demographic Characteristics. These figures show the binscatter plots between opportunity cost of time and age (Panel A) and income level (Panel B) using the grocery panel data. Opportunity cost of time is calculated by multiplying the returns to shopping (elasticity between time and money from shopping) from Aguiar and Hurst (2007a) and average spending per shopping trip.

B.2 Testing Productivity Change

TFP change in home production. I test if the unit tax has an impact on meal production productivity, which I measure using the total factor productivity (TFP). If this is the case, the tax can generate both private and social benefits at a low (or no, depending on the magnitude of the productivity increase) abatement cost.

This exercise is implemented in three steps. In Step 1, I merge datasets on food intake (output of the production function), grocery purchases and time use (two inputs of the production function) using demographic characteristics. I use demographic variables including income, family size, age, and the tax status to create cells and merge data. Then I calculate the average value for food intake, grocery purchases, and time use for each cell.

In Step 2, I estimate TFP for each cell by calculating the residual of the production function. For this, I use the factor share approach, which exploits the first order condition of cost minimization that an input's output elasticity equals the product of that input's cost share and the scale elasticity (Syverson 2004).⁴² Practically, I compute TFP based on the following equation: $TFP_c = m_c - a_q q_c - a_t t_c$ where m_c is log of food intake quantity for cell c , a_q and a_t are cost share of each input and q_c and t_c are log of input quantities. The factor share for grocery (a_q) and time (a_t) are calculated by dividing the value of each input by total production cost. To assign a dollar value for the time input, I follow Aguiar and Hurst (2007a) similar to earlier discussion in Appendix B.1.

In Step 3, I estimate a regression model $TFP_c = \beta Tax_c + \delta \mathbf{X}_c + \epsilon_c$ where c indicates each demographic cell. One caveat in this model is that there is only a single period (2009-2010) that three datasets overlap, which implies that (1) the sample size is small at 30 and (2) the tax variation is cross-sectional. The estimated β in columns (1) Appendix Table B.1 is null, suggesting that the TFP barely changed due to the food waste tax. The identical conclusion holds even if I control for income, family size, and age in column (2). This finding is not surprising given that the TFP is essentially capturing the residual in the production function. Because a 5.5% reduction in grocery

⁴²One potential drawback of the factor share approach is in its assumption that the cost of adjustment is zero (De Loecker and Syverson 2021). While this might be of a serious concern for firms substituting between labor and capital, less so is true for home production where households substitute between time and groceries, which are much more flexible than the firm setting.

Table B.1: The Effect of Unit Tax on Productivity

	(1)	(2)
Constant	0.579 (0.076)	0.718 (0.114)
Unit Tax	0.005 (0.112)	-0.007 (0.086)
Income: High		-0.044 (0.092)
Family Size: 3-4		-0.055 (0.098)
Family Size: 5+		0.082 (0.117)
Age: 30-55		0.039 (0.103)
Age: 55+		-0.427 (0.108)
N. Obs	30	30

Note:

This table reports the effect of the unit tax on meal production productivity. For this, I link three datasets on food and time usage to estimate the TFP difference between tax and no-tax group.

Column (1) does not include any controls while column (2) controls for income, family size, and age.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

purchases is accompanied by a 7.6% increase in time use, by construction there is likely to be a little room for a large TFP increase.

C Calculating the Net GHG Reduction Effect

To calculate the change in net GHG emissions from wasted food due to the unit tax, I consider change in GHG emissions from three different sources: emission reductions from avoided production; emission reductions from avoided waste treatments; and emission increases due to leakage. To quantify each effect, I use per household estimates from Table 4.1 and scale up to the national level by multiplying them with the number of total households in South Korea (19.1 million).

First, for the avoided food production, I directly take the estimated effect from Table 4.1 Panel A column (8). To produce this estimate, I first convert each row of shopping records into GHG emissions by matching each grocery item (e.g., 0.5kg of tomato) to the item-specific farm-to-kitchen GHG emissions estimates (e.g., 0.7 kg CO_2eq per 1kg of tomato) from Poore and Nemecek (2018).⁴³ Then I estimate equation (1) using log of GHG as an outcome variable. I take this approach instead of an alternative—multiplying reduced grocery purchase quantity and carbon intensity of average food basket—to allow food basket composition changes due to the unit tax. When multiplying 158 with the number of households, the national GHG reduction from the production stage is 3 million tons.

Second, for the avoided food waste treatment, I multiply reduced food waste quantity with the corresponding GHG emission intensities. Specifically, since more than 95 percent of the food waste is processed in composting or animal feed processing sites in Korea, the intensity is less than 1/3 of that of food waste in landfill at 0.19 ton CO_2eq per ton of food waste. Given the amount of food waste reduction for an average household (46kg), reduction in GHG emissions from the waste treatment stage is 0.009 ton CO_2eq per household or 0.17 million tons nationally.

Lastly, for the emission increases due to leakage, I multiply increased food waste quantity in landfill after the unit tax with the corresponding GHG emission intensities. To quantify the former, I assume that the reduced food waste (46kg) not attributable to the reduction in grocery purchases (41kg) end up in landfill. Then, I multiply this with the GHG intensity of food waste in landfill (0.655 ton CO_2eq) from the national inventory report (GGIRC 2015).⁴⁴ Given that the leakage size of 5kg per household, increased GHG emissions from leakage is 0.003 ton CO_2eq per household or 0.06 million tons nationally. Taken three sources together, the net GHG reduction effect of the unit tax is 0.163 ton CO_2eq per household or 3.1 million ton CO_2eq nationally.

⁴³Poore and Nemecek (2018) estimate the distribution of farm-to-kitchen GHG emissions for the 40 food items. Practically, I take the median GHG intensity for each item. Also, when an item is not in the list, I match it to the closest item on the list.

⁴⁴I use the “default method” which could be less accurate but allows comparison across different waste disposal methods (Hiraishi et al. 2000).

D Additional Tables and Figures

Table D.1: Effect of the Unit Tax on Food Waste Generation (Stacked DD)

	(1)	(2)
(%) Unit Tax	-0.206*** (0.044)	-0.152** (0.066)
Baseline Controls	No	Yes
Observations	3,136	3,136
Year $\times$ Stack FE	✓	✓
Municipality $\times$ Stack FE	✓	✓

This table presents the effect of the unit tax on food waste generation using the stacked DD approach. I only report coefficients for the food waste tax term, but the baseline control variables are included in column (2). All standard errors are clustered at the municipality level. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

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Table D.2: Unit Tax and Food Purchases (Alternative Specification)

	(1)	(2)	(3)	(4)	(5)	(6)
Unit Tax	-0.047** (0.020)	-0.042* (0.022)	-0.099*** (0.031)	-0.022 (0.023)	-0.052*** (0.017)	-0.057** (0.028)
Dependent Variable in Log	Grocery kg Per HH	Grocery exp Per HH	Perishable kg Per HH	Storable kg Per HH	Grocery kg per HH per Trip	GHGs Per HH
In Level	-35	-152	-28	-10	-0.23	-147
Municipality $\times$ HH ID FE	Yes	Yes	Yes	Yes	Yes	Yes
If APT $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,872	2,872	2,870	2,872	2,872	2,872

Note:

This table reports the impact of the unit tax on food purchases with a specification that allows housing type specific annual shocks. All outcome variables are in log scale. I report the coefficient of interest only, but all the regressions include baseline control variables. Standard errors are clustered at the municipality level. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

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Table D.3: The Effect of the Unit Tax on Household Meal Production Efforts (Alt Specification)

	(1)	(2)	(3)	(4)	(5)	(6)
Unit Tax Group x Post	7.60*** (2.67)	5.80*** (1.40)	1.65 (1.00)	0.123 (0.125)	0.027 (1.01)	0.382** (0.185)
Dependent Variable	Overall Time	Prep Time	Cleanup Time	Diary Time	Shopping Time	Dollars Per Day
Province $\times$ Housing Type FE	Yes	Yes	Yes	Yes	Yes	Yes
Survey Date FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	82,348	82,348	82,348	82,348	82,348	49,531

Note:

This table reports the effect of the unit tax on household meal production effort using household level data. A binary unit tax group status is assigned based on the fraction of households subject to the unit tax in 2009 (1 if the ratio is over 50

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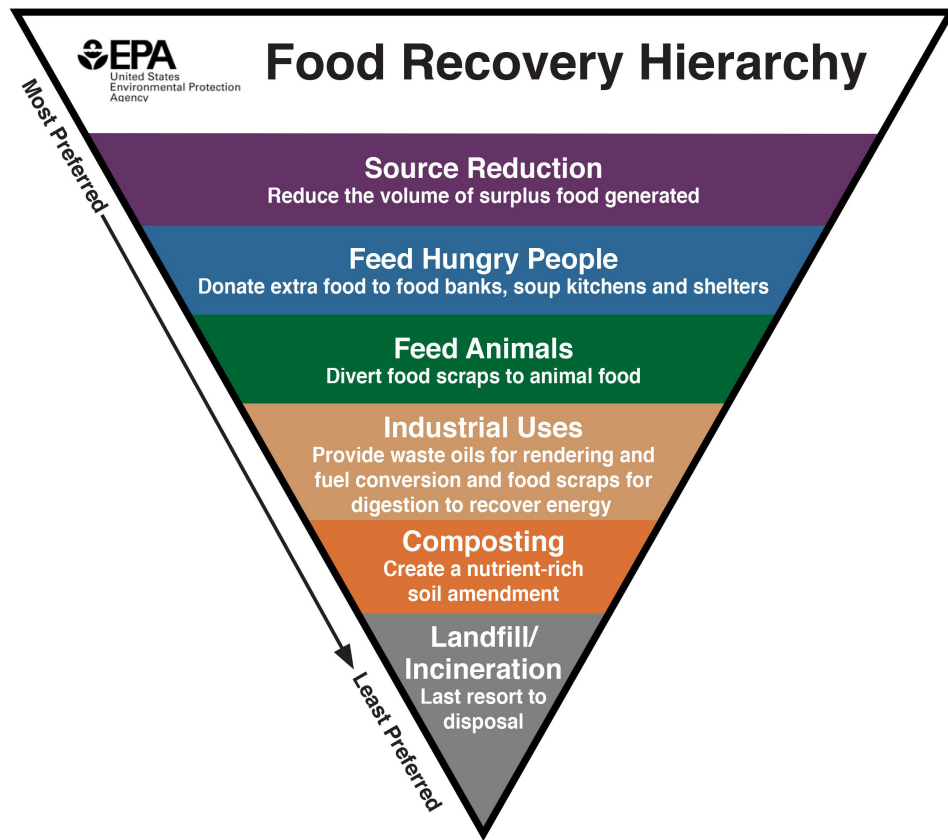
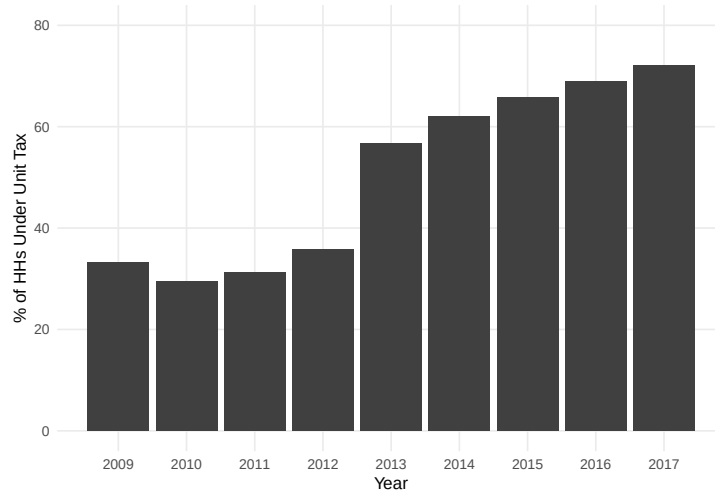
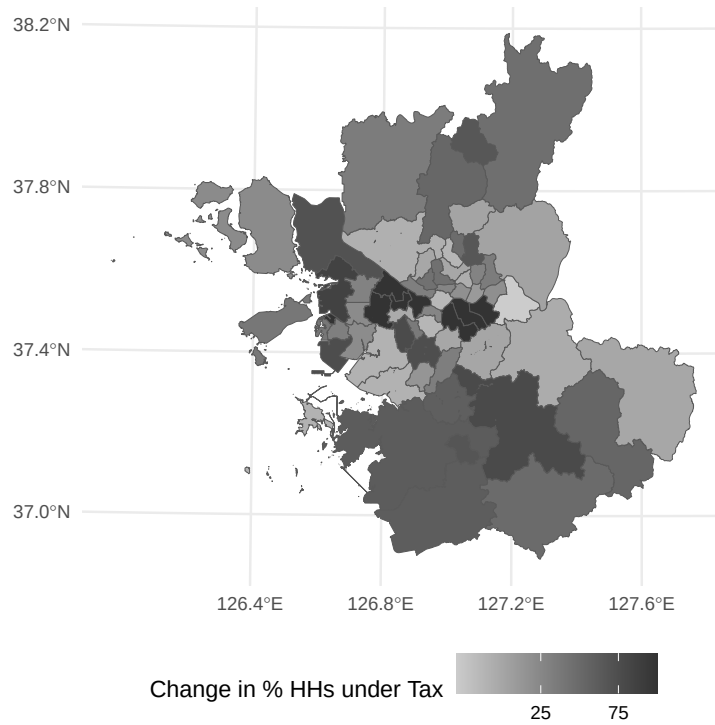


Figure D.1: Food Recovery Hierarchy

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(a) Average % of Households under the Unit Tax



(b) Change in % of HHs under Tax between 2009-17 by Municipality

Figure D.2: Wave 2 Expansion. Panel (a) shows the fraction of households subject to the food waste tax between 2009 to 2017 from 63 municipalities in the metropolitan Seoul area. A corresponding map is produced in Panel (b).

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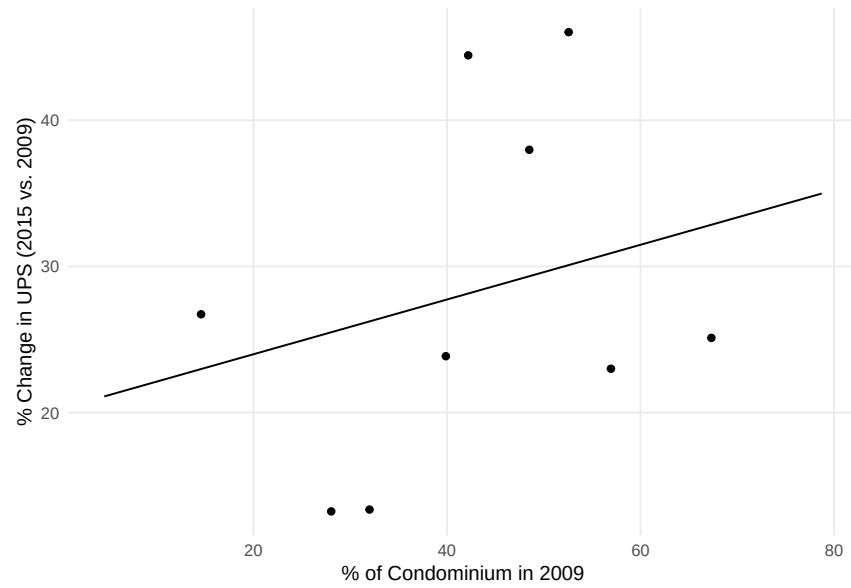


Figure D.3: This figure shows the relationship between the proportion of households living in condos in 2009 and the change in the proportion of households under the tax between 2009 and 2015 using binned regression.

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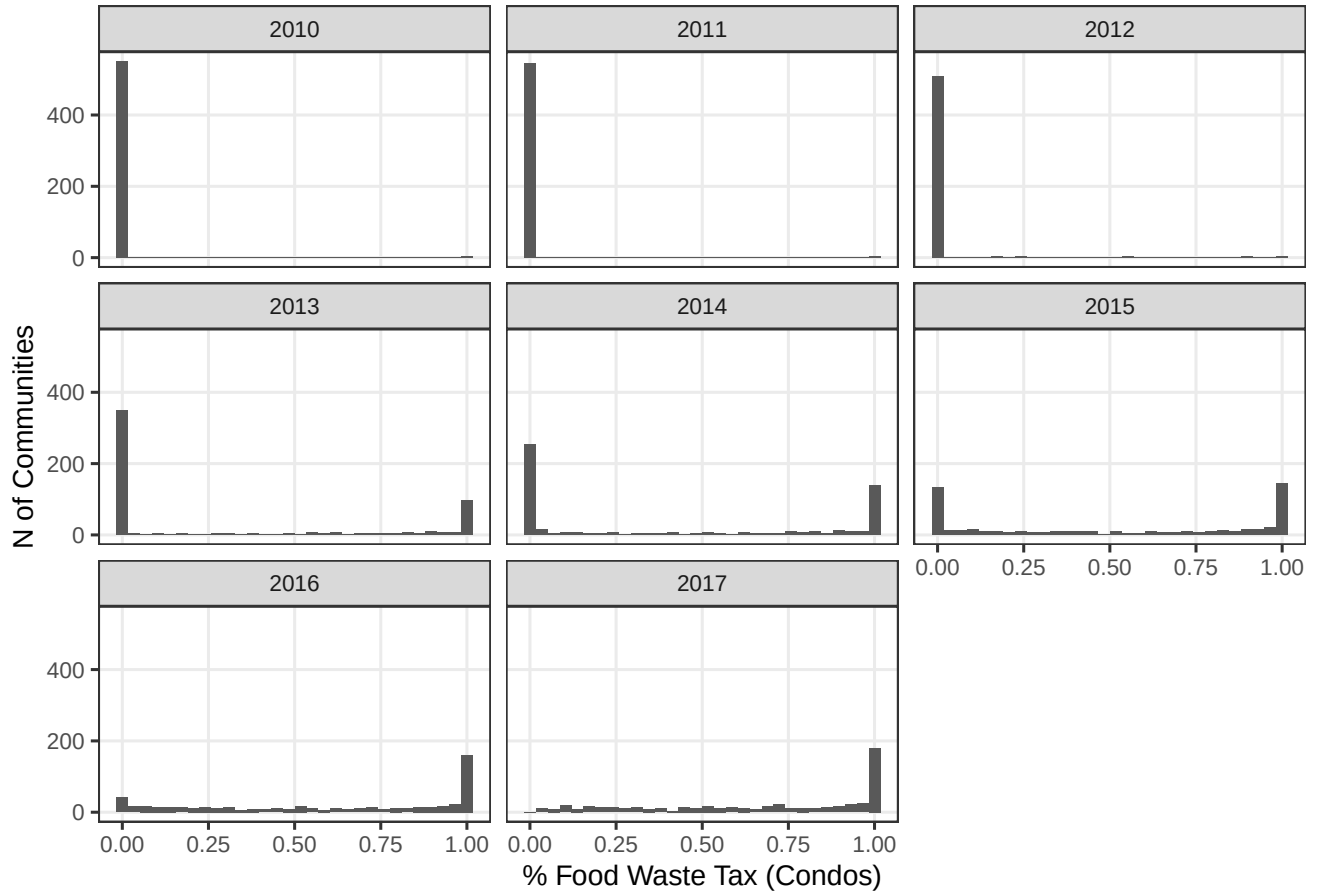


Figure D.4: Distribution of the Fraction of Condominium Residents Under the Tax at the Community Level for 2010-2017. These panels illustrate the distribution of the fraction of condominium residents under the food waste tax at the community (the smallest administrative unit in South Korea) level for years 2010-2017. The data is from 1028 communities within three provinces in the metropolitan Seoul area.

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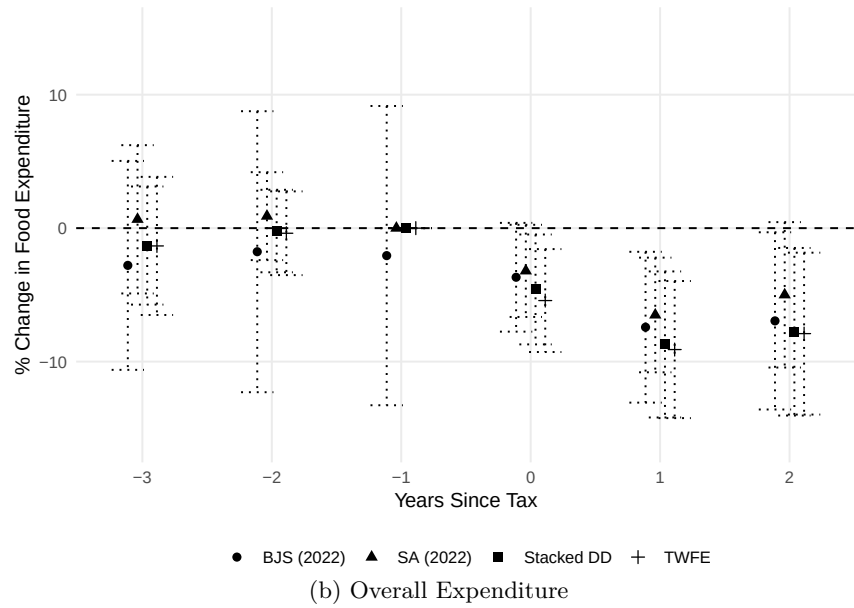
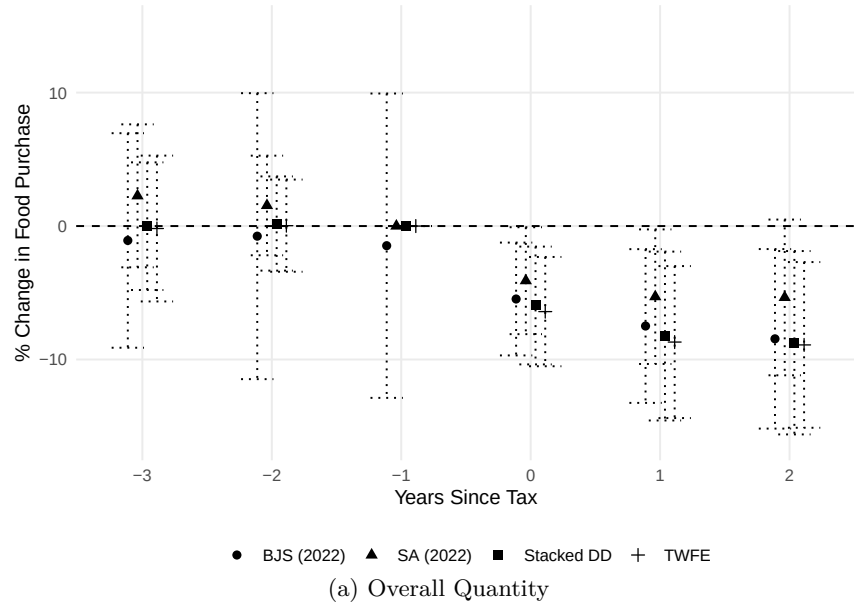


Figure D.5: The Effect of the Food Waste Tax on Grocery Purchases. This figure overlays event study style plots estimated from five different methods: a dynamic TWFE model, a stacked difference-in-difference model a la Cengiz et al., (2019), Sun and Abraham (2021) estimator, Callaway and SantAnna (2021) estimator, and Borusyak et al., (2021) estimator. The outcome variable is (a) log of per household annual grocery purchase (in kg) and (b) log of per household annual grocery spending (in USD). Event time is defined relative to the treatment year, namely the first year a household is subject to the food waste tax. I impose endpoint restrictions to estimate the effect using a nearly balanced panel. The bars represent 95 percent confidence intervals. Standard errors are clustered at the municipality level.

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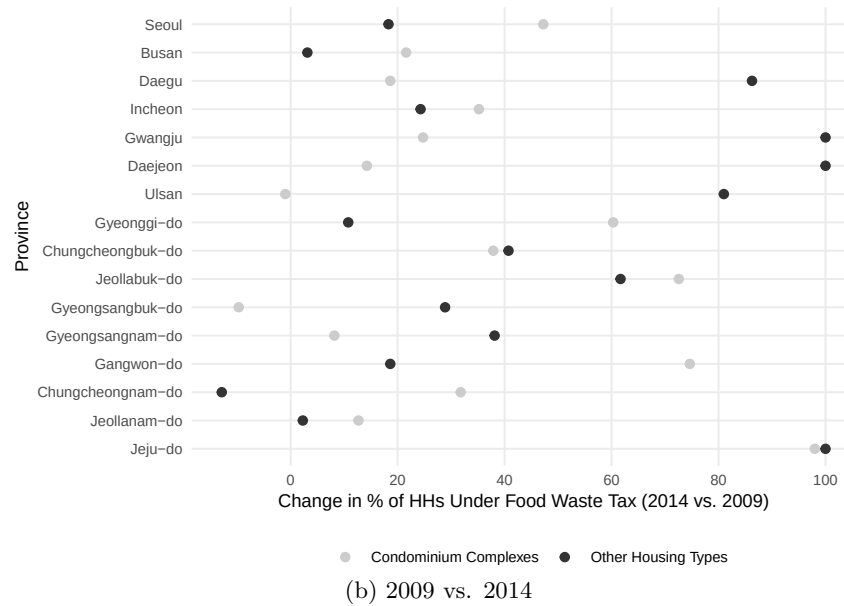
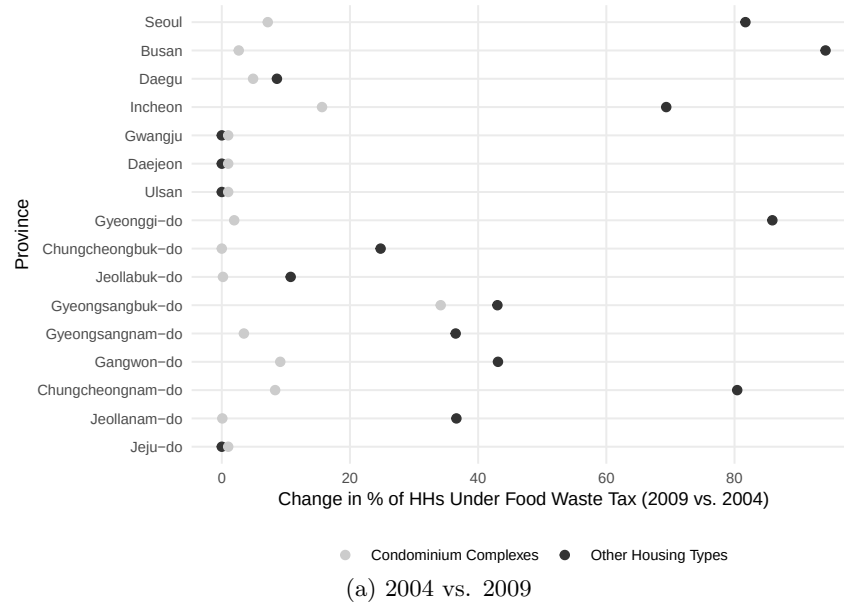


Figure D.6: Change in the Proportion of Households under Food Waste Tax at the Province Level. These figures show the proportion of households under the food waste tax at the province level for two different housing types. Panel (a) illustrates the change between 2004 and 2009 (the Wave 1 effect) while panel (b) is the change between 2009 and 2014 (the Wave 2 effect).

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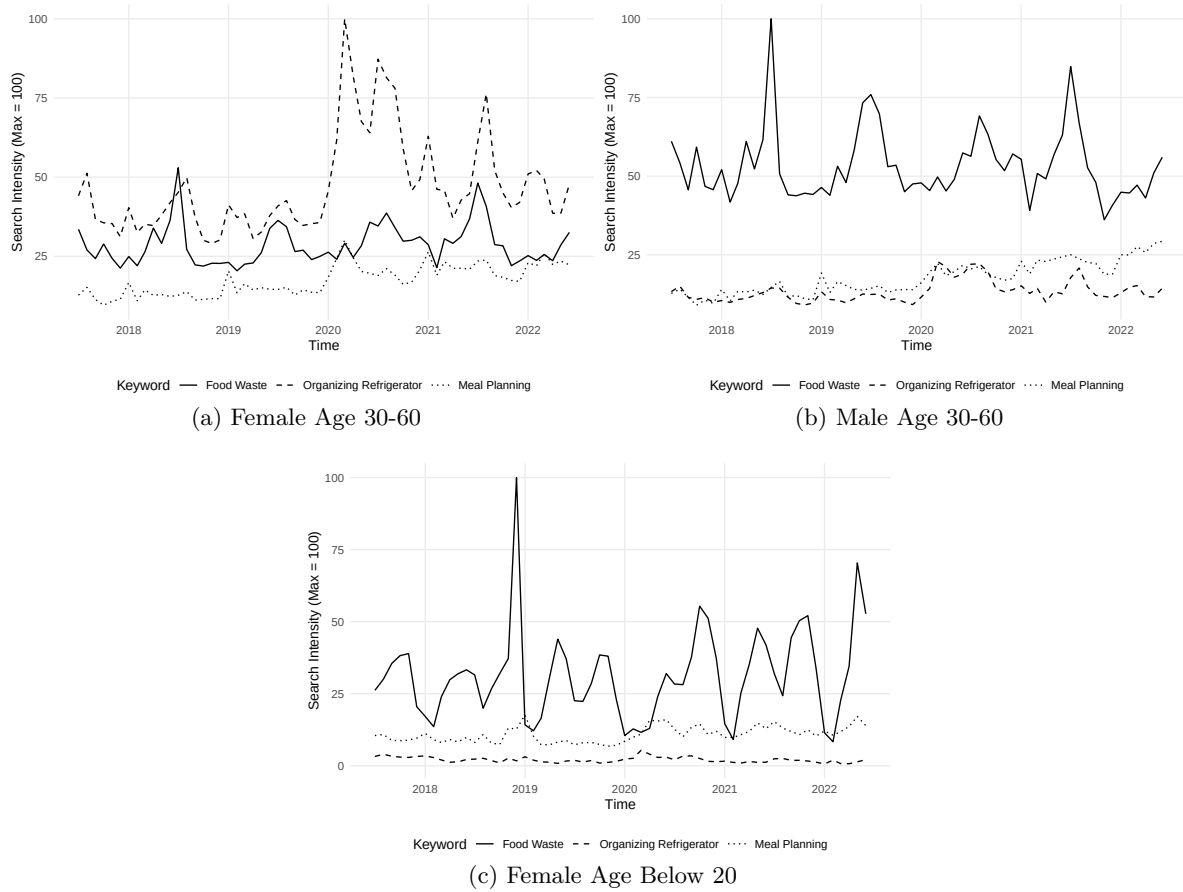


Figure D.7: Trends in Internet Search Keywords. These figures show the trend in web search intensity for three food waste related keywords (food waste, organizing refrigerator, and meal planning) for different population groups between July 2017 and Jun 2022 from Naver, a dominant search engine in South Korea. Y-axis has been normalized based on the maximum search intensity over the five years period for each demographic group. Panel (a) is for female age between 30-60, panel (b) is for male age between 30-60, and panel (c) is for female age below 20.

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